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FLASH radiation reprograms lipid metabolism and macrophage immunity and sensitizes medulloblastoma to CAR-T cell therapy

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Abstract

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Author contributions

H.N. performed the experiments, analyzed the results and produced the figures. Z.J.R. designed the experiments. W.Z. and S.A.O.M. conducted the radiation dosimetry and delivery. M.N.A. and R.M. helped with the T cell assays. R.P. and M.H. contributed to the initial in vivo RT experiments. D.Z. and L.Z. contributed to the bioinformatic analysis. H.Z., R.Z. and G.N. helped with the experimental assays. J.B.F. contributed to the GD2 CAR-T treatment. E.S.D., M.M.K., A.M., J.F.D., J.M. and C.K. contributed to the FLASH physics and RT. D.G.K., Y.G. and Y.F. contributed to the experimental design and data interpretation. D.G.K., Y.G. and Y.F. supervised the project. Y.F. conceptualized the ideas and wrote the manuscript. All authors commented on the manuscript.

Competing interests

D.G.K. is a cofounder of and stockholder in XRAD Therapeutics, which is developing radiosensitizers. D.G.K. is a member of the scientific advisory board and owns stock in Lumicell, a company commercializing intraoperative imaging technology. None of these affiliations represent a conflict of interest with respect to the work of this manuscript. D.G.K. is a coinventor on a patent for a handheld imaging device and is a coinventor on a patent for radiosensitizers. None of these patents are relevant to this manuscript. XRAD Therapeutics, Merck, Bristol Myers Squibb and Varian Medical Systems have provided research support to D.G.K. but this did not support the research described in this manuscript. Z.J.R. is listed as an inventor for intellectual property related to genetic testing for brain tumors that is managed by Duke Office of Licensing and Ventures, which is not relevant to this manuscript. The other authors declare no competing interests.

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FLASH radiotherapy holds promise for treating solid tumors given the potential lower toxicity in normal tissues but its therapeutic effects on tumor immunity remain largely unknown. Using a genetically engineered mouse model of medulloblastoma, we show that FLASH radiation stimulates proinflammatory polarization in tumor macrophages. Single-cell transcriptome analysis shows that FLASH proton beam radiation skews macrophages toward proinflammatory phenotypes and increases T cell infiltration. Furthermore, FLASH radiation reduces peroxisome proliferator-activated receptor- γ (PPAR γ) and arginase 1 expression and inhibits immunosuppressive macrophage polarization under stimulus-inducible conditions. Mechanistically, FLASH radiation abrogates lipid oxidase expression and oxidized low-density lipid generation to reduce PPAR γ activity, while standard radiation induces reactive oxygen species-dependent PPAR γ activation in macrophages. Notably, FLASH radiotherapy improves infiltration and activation of chimeric antigen receptor (CAR) T cells and sensitizes medulloblastoma to GD2 CAR-T cell therapy. Thus, FLASH radiotherapy reprograms macrophage lipid metabolism to reverse tumor immunosuppression. Combination FLASH–CAR radioimmunotherapy may offer exciting opportunities for solid tumor treatment.

Childhood cancer is a leading cause of death in children. Among these, medulloblastoma (MB) is the most frequently occurring malignant brain tumor in children¹. Despite aggressive treatments involving surgical resection, standard ionizing irradiation and chemotherapy, the prognosis for persons with high-risk MB remains poor². Immunotherapy holds great promise in improving brain cancer outcomes; however, current immunotherapy methods, which primarily focus on T cell use or activation, face notable challenges in treating brain tumors. These challenges arise mainly from an immune-hostile microenvironment that hinders T cell infiltration and activation within the tumors. Because of their immunologically inert characteristics, most brain tumors display resistance to T cell-based immunotherapies, including checkpoint blockade and adoptive cell transfer using chimeric antigen receptor (CAR)-modified T cells^{3–6}. In brain tumors, the primary source of immunosuppression originates from tumor-associated myeloid cells, particularly macrophages (M ϕ s) that constitute the majority of non-neoplastic cells^{7,8}. Tumor-associated M ϕ s usually undergo alternative M2 polarization and secrete immunosuppressive factors, such as interleukin 10 (IL-10), transforming growth factor- β (TGF β) and arginase 1, to induce tumor immunosuppression^{9–15}. Consequently, the development of promising approaches that reprogram M ϕ s to overcome tumor resistance to immunotherapy is critically needed for the treatment of pediatric brain tumors.

Radiotherapy (RT) has been generally considered as an ‘in situ vaccination’ treatment to stimulate antitumor immunity as it causes tumor cell lysis to release tumor-specific antigens that can be recognized by immune cells¹⁶. However, the impact of standard radiation on M ϕ phenotypes appears to be conflicting, depending on dose, location and cancer types^{17–23}, posing challenges for combining immunotherapy with standard RT. Notably, growing evidence suggests that ultrahigh-dose-rate delivery of radiation (that is, FLASH RT at a dose rate of ≥ 40 Gy s⁻¹) can improve the therapeutic ratio of radiation and reduce normal tissue toxicity^{24–28}. Considering the crucial importance of preserving normal brain functions and neurocognitive benefits²⁹, FLASH RT holds great promise for treating the pediatric population with brain tumors. Here, we sought to investigate the therapeutic

effects of FLASH RT on MB immunity and the underlying regulatory mechanism through single-cell and bulk transcriptome analyses. Our data show that FLASH RT stimulates proinflammatory M ϕ phenotypes in vitro and in vivo through regulating reactive oxygen species (ROS)-dependent peroxisome proliferator-activated receptor- γ (PPAR γ) activation. Strikingly, FLASH RT robustly improves CAR-T cell infiltration into MB tumors. Our study suggests that combined FLASH and CAR-T radioimmunotherapy is a promising strategy for treating pediatric brain tumors.

Results

Standard and FLASH RT improves survival in MB-bearing mice

To explore the effects of radiation in MB, we took advantage of a genetically engineered mouse MB model, based on Math1-Cre-driven specific overexpression of Smo^{W539L} (that is, SmoM2) in cerebellar granule cell neuron precursors in the hind brain (Fig. 1a). This model recapitulates the key features of the human sonic hedgehog type of MB, the most common subtype in young children under 3 years of age. Using a small animal radiation research platform incorporating high-resolution computed tomography, we were able to stereotactically deliver proton beams into mouse MB tumors at a standard (0.7 Gy s⁻¹) or FLASH (~100 Gy s⁻¹) dose rate (Fig. 1a). Our results show that 10-Gy FLASH and standard RT significantly ($P < 0.05$) and equivalently extended animal survival by ~50%, providing comparable tumor control (Fig. 1b).

FLASH RT stimulates antitumor immunity in MB

We next analyzed the impact of RT on global transcriptome in different cell populations using single-cell RNA sequencing (RNAseq). Nonlinear dimensionality reduction by uniform manifold approximation and projection (UMAP) analysis of the whole transcriptome gene signature assigned the single cells into several transcriptionally distinct cell clusters, including tumor cells, endothelial cells, oligodendrocytes/astrocytes and microglia/leukocytes (Fig. 1c,d). Interestingly, transcriptome analysis revealed that the top genes upregulated in FLASH-irradiated tumors included CD8a, a marker of cytotoxic T cells, and CD80 and CD86, two costimulatory receptors that are critical for T cell-activating responses in myeloid cells (Fig. 1e), suggesting a potential antitumor proinflammatory effect by FLASH RT. Consistent with these findings, analysis of CD11b⁺ myeloid cells showed that FLASH RT induced more robust expression of CD86 and CD80 and less expression of CD206, a surface marker of immunosuppressive myeloid cells (Fig. 1f). Further analysis of these CD11b⁺ myeloid cells identified *Arg1* (arginase 1) among the most downregulated genes in FLASH-irradiated tumors (Fig. 1g,h). *Arg1* is a known immunosuppressant that inhibits T cell activity in tumors³⁰. To validate these results, we performed flow cytometry analysis of tumor-derived single-cell suspensions. Our results showed that FLASH enhanced the infiltration of total CD45⁺ hematopoietic cells (Fig. 1i and Extended Data Fig. 1a). Interestingly, FLASH RT increased the proinflammatory M1-like CD86⁺ M ϕ population (Fig. 1j) and decreased anti-inflammatory M2-like CD206⁺ M ϕ population (Fig. 1k) compared to the no RT condition, while standard RT did not robustly affect these M ϕ populations, suggesting that FLASH RT stimulates a more proinflammatory response in tumor M ϕ s. In addition, flow cytometry analysis showed that FLASH or standard RT did

not affect the population of total M ϕ s or total and M1-like microglia cells (Extended Data Fig. 1b,c) and only FLASH RT reduced the population of M2-like microglia cells (Extended Data Fig. 1d). In accordance with these findings, FLASH RT enhanced the infiltration of CD3⁺ T cells into the tumors (Fig. 1l). Moreover, FLASH RT increased the percentage of CD8⁺ T cells in total T cells and the ratio of CD8⁺ to CD4⁺ T cells (Fig. 1m–o). FLASH and standard RT slightly yet insignificantly ($P > 0.05$) increased the infiltration of natural killer (NK) cells (Extended Data Fig. 1e). Taken together, these findings indicate that FLASH RT induces more favorable immune responses in MB compared to standard RT.

FLASH RT induces proinflammatory M ϕ phenotypes in vitro

We next investigated the effects of radiation on M ϕ polarization, a key cellular process that regulates M ϕ function and immunosuppression. Mouse bone marrow (BM)-derived M ϕ s were irradiated by a 5-Gy standard and FLASH proton beam, followed by M1-like and M2-like induction with lipopolysaccharide (LPS) and IL-4, respectively (Fig. 2a). Interestingly, FLASH RT enhanced the population of CD80⁺ proinflammatory M ϕ s under the M1 condition (Fig. 2b). Reverse transcription (RT)–PCR analysis verified that FLASH RT enhanced the expression of proinflammatory IL-1 β , while standard RT reduced its expression (Fig. 2c). Similar results were observed in irradiated human peripheral blood mononuclear cell (PBMC)-derived M ϕ s (Extended Data Fig. 2). In accordance with these data, ELISA assays showed that FLASH RT but not standard RT enhanced the expression of proinflammatory cytokines including IL-1 β and tumor necrosis factor (TNF) in these M1-like M ϕ s (Fig. 2d,e). Furthermore, FLASH or standard RT did not significantly ($P > 0.05$) affect the population of anti-inflammatory CD206⁺ M ϕ s (Fig. 2f). However, FLASH but not standard RT reduced the expression of immunosuppressive arginase 1 (Fig. 2g). To test the effect of RT on M ϕ effector functions, control or irradiated M ϕ s were incubated with human T cells, followed by flow cytometry analysis for T cell proliferation and activation. Our data show that IL-4-treated M2-like M ϕ s inhibited T cell proliferation and expression of CD25, a marker of T cell activation (Extended Data Fig. 3). Importantly, FLASH RT but not standard RT rescued T cell proliferation (Extended Data Fig. 3a) and CD25 expression (Extended Data Fig. 3b) in the T cells incubated with these M ϕ s. Together, these findings suggest that FLASH RT stimulates M ϕ s toward proinflammatory M1 polarization and leads to T cell activation.

FLASH RT induces less PPAR γ expression and M ϕ suppression

To investigate the molecular mechanism by which FLASH and standard radiation modulate M ϕ function, we performed bulk RNAseq analysis of irradiated M ϕ s under control or M1-like or M2-like condition. Our transcriptome analysis showed that LPS and IL-4 markedly switched global expression profile under unirradiated conditions (Fig. 3a). LPS robustly upregulated expression of CD86, while IL-4 enhanced expression of CD206 (*Mrc1*) and arginase 1 (*Arg1*) (Fig. 3b). Strikingly, radiation, particularly by standard RT, abrogated the transcriptome shift induced by LPS and IL-4 (Fig. 3a). Consistent with these findings, FLASH and standard RT seemed to evoke different immune responses, as FLASH RT induced less expression of immunosuppression-associated genes including *Mrc1* and *Arg1* and also more expression of immunostimulation-associated genes, including *Cd86*, *Il1b* (IL-1 β) and *Tnf* (TNF), particularly in the M1-stimulatory condition (Fig. 3c). In accordance

with our results from in vitro flow cytometry and RT–PCR analyses (Fig. 2), these findings collectively suggest that FLASH RT may cause a transcriptomic change toward a more proinflammatory and less anti-inflammatory status.

We explored the potential transcriptional regulation of M ϕ functions in these settings. We initially analyzed the expression of transcription factors (TFs) that are known as regulators of immunosuppressive M2 M ϕ polarization, including PPAR, Spi1, Cebp, Stat6, Klf4, Irf8, Fos, Jun and hypoxia-inducible factors HIF1 α and HIF2 β (*Epas1*). Notably, PPAR γ was identified as the most robustly downregulated and upregulated factor by LPS and IL-4, respectively, in our system, which led us to focus our next study on PPAR γ (Fig. 3d). Compared to standard RT (+51.1%, versus unirradiated cells), FLASH RT induced substantially less expression of PPAR γ (+1.8%, versus unirradiated cells) but not PPAR α or PPAR δ in M ϕ s under the M1 condition (Fig. 3e). Likewise, FLASH RT inhibited M2 condition-inducible expression of arginase 1 in M ϕ s (Fig. 3f). Considering a well-established role of PPAR γ and arginase 1 for immunosuppressive polarization of tumor M ϕ s³¹, these findings suggest that FLASH RT may inhibit PPAR γ expression to drive less immune-inhibitory and more immune-stimulative phenotypes in M ϕ s.

FLASH RT inhibits oxidized low-density lipoprotein (oxLDL) generation to reduce PPAR γ activity

To define the mechanism underlying FLASH RT-induced downregulation of PPAR γ , we initially tested its effects on ROS generation. Our results showed that FLASH RT did not induce a detectable increase in ROS production in mouse M ϕ s, while standard RT evoked a time-dependent ROS generation, peaking at 12 h after irradiation (Fig. 4a). Consistent with this result, FLASH RT induced less ROS generation in human PBMC-derived M ϕ s compared to standard RT (Extended Data Fig. 4a). Moreover, FLASH RT inhibited PPAR γ activity but standard RT increased its activity in mouse M ϕ s (Fig. 4b). Similar results were observed in irradiated human M ϕ s (Extended Data Fig. 4b). Importantly, pretreatment of M ϕ s with ROS scavenger TEMPO abrogated standard RT-induced PPAR γ activation (Fig. 4c), indicating that standard RT stimulates PPAR γ activation through ROS. Likewise, scavenging ROS reduced arginase 1 expression in both nonirradiated and irradiated cells, suggesting a requisite role for ROS in arginase 1 expression (Fig. 4d). Together, these findings suggest that FLASH RT does not affect ROS production but inhibits PPAR γ activity and standard RT stimulates ROS-dependent PPAR γ activation and arginase 1 expression.

We next explored a potential role of oxLDL that contains oxidized lipids known to activate PPAR γ ^{32,33}. Our data showed that FLASH RT inhibited oxLDL generation but standard RT enhanced it in mouse M ϕ s (Fig. 4e). Similar results were observed in irradiated human M ϕ s (Extended Data Fig. 4c). Production of oxLDL proceeds through ROS-mediated direct reaction and lipid oxidases including nicotinamide adenine dinucleotide phosphate oxidase (*Nox*), lipoxygenase (*Lox*) and myeloperoxidase (*Mpo*). Notably, both FLASH and standard RT markedly inhibited M ϕ expression of two major lipid oxidases, Alox12 and *Mpo*, as well as Alox5 to a lesser extent, as revealed by RNAseq analysis (Fig. 4f). RT–PCR analysis confirmed that FLASH and standard RT abrogated *Mpo* mRNA expression (Fig.

4g). This is likely because of a global altered activation of multiple TFs (Extended Data Fig. 5). Furthermore, pretreatment of M ϕ s with ROS scavenger TEMPO inhibited standard RT-stimulated oxLDL production (Fig. 4h). These findings collectively suggest that FLASH radiation may reduce PPAR γ activity through downregulation of oxidase expression, while standard radiation may enhance PPAR γ activation through ROS in M ϕ s.

FLASH RT sensitizes tumors to GD2 CAR-T cell therapy

Because FLASH RT stimulates proinflammatory polarization in M ϕ s in vitro and in vivo (Figs. 1–3) and enhances T cell infiltration into MB tumors in vivo (Fig. 1), we hypothesize that FLASH RT may sensitize brain tumors to T cell-based immunotherapy. To test this hypothesis, we aimed to develop immunotherapy using CAR-engineered mouse T cells that specifically target GD2, a well-known target for brain tumors^{34,35}. To validate the expression of this target, we performed immunofluorescence and flow cytometry analyses of the tumors derived from our genetically engineered MB model. Tissue immunofluorescence analysis of mouse normal brain and MB tumors verified that GD2 was specifically and robustly expressed in MB tumors (Fig. 5a), which was further validated by flow cytometry analysis showing more than 60% of tumor-derived cells were GD2⁺ (Fig. 5b), suggesting that GD2 is a vital and selective therapeutic target for MB. In addition, using a retrovirus-mediated murine CAR-T system we previously developed^{36,37}, we generated murine GD2 CAR-T cells with >40% CAR⁺ T cells (Fig. 5c). An in vitro killing assay with tumor cells indicated that these CAR-T cells could directly induce cytotoxicity in GD2⁺ mouse brain tumor cells in vitro (Fig. 5d). We investigated the effects of GD2 CAR-T cell monotherapy on animal survival in MB-bearing mice. MB was genetically induced in mice, followed by treatment with GD2 CAR-T cells injected intravenously (Fig. 5e). Unexpectedly, GD2 CAR-T cell therapy did not improve animal survival (Fig. 5f), likely because of no detectable tumor infiltration by CAR-T cells after infusion (Fig. 5g).

To test the effects of FLASH RT on CAR-T cell infiltration and animal survival, MB-bearing mice were treated by standard and FLASH RT, followed by T cell therapy with GD2 CAR or control single-chain variable fragment (scFV)-free CAR-T cells 5 days after the irradiation (Fig. 6a). Our data showed that combination treatment with FLASH RT plus GD2 CAR-T cells substantially extended animal survival (+27.5 days median, $P < 0.001$, compared to FLASH RT plus control CAR-T cells), while combination treatment with standard RT plus GD2 CAR-T cells moderately extended animal survival (+11.0 days median, $P < 0.001$, compared to standard RT plus control CAR-T cells) (Fig. 6b). Of note, 70% of the mice that received FLASH RT plus GD2 CAR-T combination therapy remained alive, whereas all mice in other groups died 83 days after tumor induction (Fig. 6b). In addition, we tested the effects of the combination therapy in a syngeneic glioma mouse model (Extended Data Fig. 6a). Our data showed that FLASH RT monotherapy alone induced a robust therapeutic effect (+29.5 days median, $P < 0.0001$, compared to control nonirradiated mice), while standard RT therapy induced a moderate effect (+10.5 days median, $P < 0.01$, compared to control nonirradiated mice) (Extended Data Fig. 6b). Strikingly, combination of FLASH RT with GD2 CAR-T cell therapy markedly improved animal survival, with 80% of the mice remaining alive when experiments reached the endpoint on day 60 (Extended Data Fig. 6b). The combination therapy also substantially inhibited tumor growth (Extended Data Fig. 6c).

We determined the effects of FLASH RT on CAR-T cell infiltration and function in vivo. Whole-body bioluminescence imaging showed that FLASH RT but not standard RT robustly improved the homing of these nanoluciferase (nLuc)-expressing CAR-T cells to the MB tumors (Fig. 6c), likely contributing to the survival benefits in the group of mice that received FLASH RT plus GD2 CAR-T combination therapy. Furthermore, our flow cytometry analysis of the MB tumors excised 3 days after CAR-T cell therapy showed that FLASH RT and standard RT stimulated the infiltration of total hematopoietic cells into the tumor at a similar level (Fig. 6d); however, FLASH RT but not standard RT markedly enhanced CAR-T cell infiltration (Fig. 6e). Moreover, compared to standard RT, FLASH RT more robustly stimulated expression of interferon- γ (IFN γ), a major T cell cytotoxic cytokine, and induced less reduction of expression of Ki67, a proliferative marker, in the infiltrated CAR-T cells (Fig. 6f). Furthermore, FLASH RT did not affect expression of programmed cell death protein 1 (PD1) and Tim3, two T cell exhaustion-associated markers, in the infiltrated CAR-T cells (Fig. 6g). In addition, we analyzed the tumors excised 7 days after CAR-T cell therapy (Extended Data Fig. 7a). Similarly, our data showed enhanced CAR-T infiltration by FLASH RT but not by standard RT at that time (Extended Data Fig. 7b). Moreover, FLASH RT rather than standard RT stimulated IFN γ expression in the CAR-T cells (Extended Data Fig. 7c, left). Both FLASH RT and standard RT inhibited PD1 and Tim3 expression in the CAR-T cells (Extended Data Fig. 7c, right), suggesting a time-dependent effect of combination therapy on T cell functions. In a parallel study, we tested a different therapeutic window for the RT and CAR-T cell combination therapy, in which mice were treated with GD2 CAR-T cells 3 days after the irradiation, followed by tumor analysis 6 days later (Extended Data Fig. 7d). Our data showed that FLASH RT but not standard RT consistently stimulated CAR-T cell infiltration (Extended Data Fig. 7e). Neither FLASH RT nor standard RT affected granzyme B (GranzB) or Ki67 expression in the CAR-T cells but FLASH RT stimulated more IFN γ expression in the CAR-T cells than standard RT (Extended Data Fig. 7f). Furthermore, neither FLASH RT nor standard RT affected PD1 expression but standard RT rather than FLASH RT increased Tim3 expression (Extended Data Fig. 7g). Together, these findings suggest that FLASH RT improves CAR-T cell infiltration, stimulates more favorable T cell functions and overcomes tumor resistance to GD2 CAR-T cell immunotherapy.

In summary, our work shows that FLASH radiation stimulates proinflammatory M1-like polarization of tumor M ϕ s in vitro and in vivo and enhances infiltration of endogenous T cells or infused CAR-T cells into MB tumors. We reveal a redox-mediated and PPAR γ -mediated mechanism by which FLASH RT may reprogram M ϕ s to overcome tumor resistance to T cell-based immunotherapy (Fig. 6h).

Discussion

FLASH RT, albeit initially proposed in the 1960s^{38,39}, is an innovative treatment approach given recent discoveries showing promise in the treatment of solid tumors because of its potential for reduced toxicity in normal tissues^{24–28}. However, its therapeutic effects on tumor immunity, particularly on M ϕ functions, remain largely unclear. Here, we report that FLASH RT modulates lipid metabolism to reprogram tumor-associated M ϕ s toward proinflammatory polarization and reduces M ϕ -mediated tumor immunosuppression,

empowering CAR-T cell infiltration into the tumors and sensitizing autochthonous MB to CAR-T cell immunotherapy. Because we used a primary mouse model of cancer that coevolves with the immune system, we investigated the impact of FLASH RT in a native tumor immune microenvironment, which is not subject to the influence of tumor cell transplantation that alters the response to the combination of RT and immunotherapy⁴⁰. Consistent with our findings, a recent report showed that FLASH RT increases antitumor Mφs and reduce protumor Mφs in lung cancer⁴¹. FLASH RT is, therefore, a promising combination partner for cancer immunotherapy, particularly in the context of primary brain tumors, considering its additional benefits of reduced neurocognitive side effects^{29,42}.

Our work reveals that FLASH and standard RT induces PPARγ inactivation and activation, respectively. A critical and requisite role has been well established for PPARγ in M2 polarization of tumor Mφs^{31,43}. PPARγ is a ligand-activated TF that is essential for the regulation of metabolism, inflammation, proliferation and differentiation and can be activated by naturally occurring ligands (for example, polyunsaturated fatty acids) and pharmacologically synthesized agents (for example, rosiglitazone)⁴⁴. Our initial study showed that ROS scavenger TEMPO inhibits PPARγ activity and arginase 1 expression in standard RT-irradiated Mφs, indicating that standard RT-evoked PPARγ activation depends on redox stress. Furthermore, we found that FLASH RT decreases oxLDL production while standard RT increases it, implying that FLASH RT may disrupt oxLDL formation to reduce PPARγ-mediated Mφ immunosuppression. Supporting this hypothesis, 9-HODE and 13-HODE, two major oxidized lipid components of oxLDL, can directly bind and activate PPARγ^{45–47}; a structural study validated the interaction of PPARγ with oxidized fatty acids, including 4-HDHA, 5-HEPA and 6-HOTE, leading to greater activation of PPARγ than unoxidized forms³². However, as an established marker for cardiovascular disease, the circulating oxLDL, which is favored by polyunsaturated fatty acids rather than monounsaturated fatty acids³³, stimulates proinflammatory responses by binding to its cellular surface receptors, suggesting that oxLDL and PPARγ may have different roles in cancer and cardiovascular diseases.

Fatty acid oxidation can proceed through either lipid oxidases or ROS-mediated direct reaction. Strikingly, our study showed that FLASH RT reduces production of oxLDL, which consists of oxidated cholesterol, fatty acid and protein components, likely acting primarily through downregulation of oxidases including Alox12 and MPO, as FLASH does not robustly stimulate ROS generation in Mφs. Despite inducing similar downregulation of these oxidases, standard RT does not decrease oxLDL production, possibly because of robustly induced ROS. The precise mechanism underlying the oxidase downregulation remains obscure but may be related to an adaptive gene transcription response after RT-induced mitochondrial damage and subsequent lipid metabolism dysfunction. In accordance with our findings showing that FLASH RT does not robustly stimulate ROS production, growing evidence suggests that FLASH RT generates fewer ROS in various cells than standard RT^{29,48}. The potential mechanisms may involve different responses in oxygen depletion^{29,49–51} or in potential mitochondrial dysfunction induced by FLASH and standard RT. In addition, considering that activated Mφs produce ROS to combat foreign substances, Mφs can generally tolerate a higher level of ROS to maintain redox hemostasis, likely because of the possibility that Mφs may possess a greater reserve capacity for the enzymatic

reduction of ROS and can, therefore, remove them more rapidly, potentially contributing to the minimal effects of FLASH RT on intracellular ROS level in M ϕ s.

T cell-based immunotherapies including checkpoint inhibition and adoptive cell transfer with CAR-modified T cells currently show efficacy in a minority of persons with solid tumors. This is in part because of an immune-hostile microenvironment that limits T cell infiltration into and activation in the tumor. In particular, immunologically cold tumors such as the majority of primary brain tumors are characterized by very few or no infiltrating T cells⁴. Improving T cell recruitment into the tumors is, therefore, critically needed to enhance the success of immunotherapy. Our study shows that FLASH RT enhances GD2 CAR-T cell infiltration into MB tumors, likely because of the reversal of PPAR γ -mediated M ϕ immunosuppression. In addition, this can be attributed to the potential reduction of TGF β , a known immunosuppressive cytokine, considering that FLASH RT induces less TGF β expression in normal tissue^{25,27,52,53}. Disialoganglioside GD2 is commonly overexpressed in pediatric and adult solid tumors, such as MB, neuroblastoma and glioma^{34,35,54}. Anti-GD2 monoclonal antibodies including naxitamab and dinutuximab represent the standard of care for persons with high-risk neuroblastoma. GD2 CAR-T therapy shows robust efficacy in mouse xenograft models with human MB tumors⁵⁴ but exhibits a moderate therapeutic response in glioma and its efficacy is restricted by the immunosuppressive microenvironment^{55,56}. M ϕ s are a major cell population in primary brain tumors, serving as a key source of tumor immunosuppression and causing resistance to immunotherapy^{7–15}. We show that FLASH RT overcomes tumor resistance to GD2 CAR-T cell therapy, likely because of inhibited M ϕ -mediated immunosuppression and reduced arginase 1 release from M ϕ s. It is tempting to speculate that FLASH RT may spare tumor vasculature to allow T cell delivery into the tumor given growing evidence that FLASH RT has a less toxic effect on blood vessels in normal tissues than standard RT^{25,57,58}; however, the specific effects of FLASH RT on tumor vasculature remain to be investigated.

In sum, our study reveals an ROS-driven and oxidase-driven mechanism that contributes to FLASH RT-induced proinflammatory polarization through oxLDL and PPAR γ , providing molecular insight into FLASH RT-modulated tumor immunity. Thus, combined FLASH–CAR-T radioimmunotherapy may offer exciting opportunities for treating pediatric brain tumors and possibly other malignant solid tumors.

Methods

Ethics statement

Research conducted in this manuscript complies with all relevant ethical regulations at the University of Pennsylvania. All experiments with mice were conducted in accordance with protocols (805096 and 806643) approved by the Institutional Animal Care and Use Committee at the University of Pennsylvania. University of Pennsylvania guidelines for the proper and humane use of animals in biomedical research were followed. The protocol for human monocyte and T cell collection was reviewed and approved by the Institutional Review Board of the University of Pennsylvania, with each participant providing written informed consent.

CAR-T cells

Spleen-derived T cells were isolated from C57/B6 mice (2 months old, half male and half female, Jackson laboratory) by mechanical dissociation using a gentleMACS dissociator (Miltenyi Biotech), followed by filtration through a sterile 70- μ m strainer (NETA, 410-0002-OEM). T cells were isolated using an EasySep mouse T cell isolation kit (StemCell, 19851). T cells were cultured in Iscove's modified Dulbecco's medium with 10% FBS. T cells were treated with anti-CD3 ϵ and anti-CD28 antibodies (1 μ g ml⁻¹; BioLegend, 100302 and 102102) and recombinant IL-2 (100 IU per ml, Peprotech, 212-12) for 2 days. Before retrovirus transduction, non-tissue-treated 6-well or 12-well plates were treated with retronectin (20 μ g ml⁻¹ in PBS; Takara, T100A) at 4 °C overnight. Retrovirus expressing GFP-nLuc, mouse GD2 CAR or control scFV-free CAR sequence was prepared in Phoenix cells cotransfected with a pCL-Eco helper plasmid using Lipofectamine 2000 (Life Technologies, 11668-019)^{37,59}. Cells were immunostained with biotinylated protein L (Life Technologies, 29997) to analyze CAR expression using an FACS-Canto II flow cytometer (BD Biosciences). The flow cytometry data were analyzed using FlowJo (version 10.8.1) software. A CAR-T cell-killing assay was performed⁶⁰.

Animal model and treatment

Genetically engineered MB was induced in mice^{61–64}. In brief, *Math1-Cre;SmoM2^{fl/fl}* (Rosa-LSL-Smo^{W539L/YFP}) mice were generated by breeding *SmoM2^{fl/fl}* mice with *Math1-Cre* mice, both on a C57/B6 background. The genotype was validated by PCR analysis of genomic DNA using the following primers: Math1-Cre forward, 5'-CCGGCAGAGTTTACAGAAGC-3'; Math1-Cre reverse, 5'-ATGTTTAGCTGGCCCAAATG-3'; SmoM2^{WT} (wild type) forward, 5'-CTGGCTTCTGAGGACCG-3'; SmoM2^{WT} reverse, 5'-AGCCTGCCCAGAAGACTCC-3'; SmoM2^{mutant} forward, 5'-TCCCCATCAAGATCCATTTC-3'; SmoM2^{mutant} reverse, 5'-CTGAAGTTGTGGCCGTTTAC-3'. Mice (2 months old, half male and half female, Jackson laboratory) were administrated with CAR-T cells (5 $\times$ 10⁶ cells per mouse) through the tail vein. For T cell imaging, mice were infused with mouse T cells coexpressing GD2 CAR or control CAR with tdTomato-nLuc (3 $\times$ 10⁶ cells per mouse) through the tail vein. After retro-orbital injection of coelenterazine (10 mg kg⁻¹; Furimazine), mice were imaged. For induction of glioma in mice, 3 $\times$ 10⁵ mouse fLuc-expressing GL261 glioma cells (PerkinElmer, 134246) were orthotopically injected into the brains of WT C57BL/6 mice (6–8 weeks old, half male and half female). Survival after injection was monitored for up to 180 and 60 days for MB and glioma, respectively. Mice were injected with luciferin (150 mg kg⁻¹; GoldBio), followed by whole-body bioluminescence analysis using an IVIS 200 Spectrum imaging system to monitor tumor growth. When exhibiting severe glioblastoma symptoms including hemiparesis, dome head or more than 20% body weight loss, tumor-bearing mice were killed. Mice were randomized to receive treatment and the investigators were not blinded. All animals used in this study were housed in the animal facility accredited by the Association for the Assessment and Accreditation of Laboratory Animal Care at the University of Pennsylvania.

Stereotactic RT

A cyclotron-generated 230-MeV proton beam (IBA) was directed horizontally through scattered and collimated system for irradiation⁶⁵. The EBT3 film (Ashland Advanced Materials) was used to check the dose uniformity and alignment. The juvenile mice cerebellum region was aligned with a circular field of 8 mm in diameter and subjected to radiation at the proton beam entrance region. For the dosimetry of such a small field, a Faraday cup was used to capture the proton fluence cross-calibrated with the dose under a large uniform field with an EBT3 film and a reference Advanced Markus ion chamber (PTW Freiburg)⁶⁶. The double-scattering and collimation apparatus was dosimetrically verified with dose rates^{65,67}. The dose rate was derived from the measured dose and the irradiation duration captured from an oscilloscope (Tektronix). The proton beam can be delivered at drastically different beam current, resulting in a standard dose rate of 0.7–0.9 Gy s⁻¹ or a FLASH ultrahigh dose rate of 97–143 Gy s⁻¹. The cells were also subjected to radiation within the beam's entrance (plateau) region with a circular field of 26-mm diameter using two distinct modalities: FLASH and standard. The FLASH modality exhibited a dose rate spanning approximately 80 to 135 Gy s⁻¹, while the conventional standard modality operated within the range of 0.60 to 0.85 Gy s⁻¹. These variable dose rates were attained by manipulating the cyclotron current, which ranged from 360 nA to 2 nA for FLASH and conventional standard modalities, respectively. Mice were irradiated with the entrance (plateau) region of the beam with a field size of an 8-mm-diameter circular collimator at the hind brains. Cells cultured in the plates were irradiated. The total doses used were 10 Gy for mice and 5 Gy for cells.

Single-cell RNAseq

Treated *Math1-Cre;SmoM2^{fl/fl}* mice bearing MB tumors were killed and perfused with PBS containing EDTA. Tumor tissue was excised, followed by digestion with collagenase II (5 mg ml⁻¹; Invitrogen, 17101-015) and DNase (1 mg ml⁻¹; Sigma-Aldrich, D4527). A single-cell suspension was harvested after filtering using a mesh strainer with 70-μm pores (NETA, 410-0002-OEM). Cell samples were pooled from three mouse tumors for each group and were prepared and analyzed according to the manufacturer's V3 library protocol (10x Genomics), followed by single-cell RNAseq analysis at the Center for Applied Genomics of the Children's Hospital of Philadelphia. Reads were aligned using Cell Ranger (10x Genomics, version 6.1.2) against a mouse reference library (mouse genome assembly GRCm38/mm10). The gene expression matrices that passed default quality control metrics were integrated and analyzed by Seurat R package (version 4.0.6) using *sctransform* (version 2)^{68,69}. The downstream analysis was then conducted using Seurat⁶⁸.

Bulk RNAseq analysis

Treated mouse bone marrow-derived Mφs were lysed using TRIzol (Thermo Fisher Scientific) and RNA was isolated according to the manufacturer's instructions. RNA was purified using an RNeasy Plus mini kit (QIAGEN). DNA library was prepared with a TruSeq mRNA stranded kit (Illumina). The quality of prepared RNA and library DNA was analyzed with RNA Nano assay chips, RNA Pico assay chips and DNA Nano assay chips using a 2100 Bioanalyzer (Agilent). The library was subject to next-generation sequencing

analysis with a NovaSeq at Azenta Life Sciences. The sequences were aligned to the mouse genome assembly GRCm38/mm10 using Kallisto (version 0.46.0). The gene expression was normalized and calculated as counts per million values by R packages tximport (version 1.30.0) and EdgeR (version 3.14.0).

Mouse and human M ϕ isolation and treatment

Mouse bone marrow-derived M ϕ s were isolated^{31,70}. Freshly isolated femur and tibia bones from C57BL/6 mice (6–8 weeks old, half male and half female) were flushed with RPMI-1640 culture medium (Life Technologies). Cells were harvested and passed through a 40- μ m strainer. ACK lysis buffer (Thermo Fisher Scientific) was used to deplete red cells. Bone marrow cells were cultured in RPMI-1640 medium containing 5% FBS (Life Technologies). Cells were treated with 10 ng ml⁻¹ mouse colony-stimulating factor 1 (CSF1; PeproTech, 315-02) for 7 days to induce M ϕ differentiation, followed by treatment with 100 ng ml⁻¹ LPS (Sigma-Aldrich, LPS25) or 20 ng ml⁻¹ IL-4 (BioLegend, 574302) for 2 days with or without pretreatment with 1 mM TEMPO (Targetmol, T5363) or DMTU (Targetmol, T40615). Human PBMC-derived monocytes were isolated from the healthy volunteers, aged 16–64, and provided by the Human Immunology Core of the University of Pennsylvania. The monocytes were treated with human CSF1 (10 ng ml⁻¹; BioLegend, 574806) in RPMI-1640 medium for 5 days to induce differentiation into M ϕ s, followed by treatment with LPS (100 ng ml⁻¹; Sigma-Aldrich, LPS25) or human IL-4 (20 ng ml⁻¹; Peprotech, 200-04). The treated M ϕ s were also cocultured with human T cells (isolated from healthy volunteers and provided by Human Immunology Core of the University of Pennsylvania). For CFSE staining in T cells, 5 $\times$ 10⁷ T cells per ml were incubated with 5 μ M CFSE (BioLegend, 423801) for 20 min at 37 °C. The staining was quenched and the treated cells were analyzed.

Flow cytometry

Mouse tumor-derived single-cell suspensions, mouse bone marrow-derived M ϕ s and human PBMC-derived M ϕ s were immunostained with fluorescent dye-conjugated antibodies to CD3 (1:100; BioLegend, 100203/100233), CD4 (1:100; BioLegend, 100540), CD8a (1:100; BioLegend, 100706/100708/100733), CD45 (1:100; BioLegend, 103133/103134), F4/80 (1:100; BioLegend, 123107), CD206 (1:100; BioLegend, 141719), CD80 (1:100; BioLegend, 104713, 375403), CD86 (1:100; BioLegend, 105005; 1:100, Miltenyi Biotec, 130-102-604), IFN γ (1:100; BioLegend, 505825), GranzB (1:100; BioLegend, 372211), Ki67 (1:100; Thermo Fisher Scientific, 17-5698-80), LAG3 (1:100; eBioscience, 11-2231-80; BioLegend, 125225/125209), PD1 (1:100; BioLegend, 135223), Tim3 (1:100; BioLegend, 134009), CD11b (1:100; BioLegend, 101206/101212), CD206 (1:100; BioLegend, 321109), NK1.1 (1:100; BioLegend, 156505), TMEM119 (1:100; Thermo Fisher Scientific, 25-6119-80), CD25 (1:100; BioLegend, 101915/302610), GD2 (1:100; BioLegend, 357324) or control IgG⁷¹. For intracellular staining, cells were treated with TF staining buffer (Thermo Fisher Scientific, 00-5523-00). Cells were analyzed using an FACSCanto II flow cytometer (BD Biosciences) and FlowJo software (version 10.8.1).

Real-time RT-PCR analysis

RNA was extracted from mouse M ϕ s using the RNeasy Plus mini kit (QIAGEN, 74136) according to the manufacturer's instructions. Real-time RT-PCR was conducted using the Superscript III first-strand synthesis system (Thermo Fisher Scientific, 12574026) and the powerSYBR Green PCR master mix (Applied Biosystems, 4367659), followed by analysis a QuantStudio 6 Flex system (Applied Biosystems). The primers used were as follows: Arg1 forward, 5'-CTCCAAGCCAAAGTCCTTAGAG-3', Arg1 reverse, 5'-AGGAGCTGTCATTAGGGACATC-3', IL-1 β forward, 5'-GCAACTGTTCTGAACTCAACT-3', IL-1 β reverse, 5'-ATCTTTTGGGGTCCGTCAACT-3', Mpo forward, 5'-AGTTGTGCTGAGCTGTATGGA-3'; MPO reverse, 5'-CGGCTGCTTGAAGTAAACAGG-3'.

Detection of ROS

Treated mouse bone marrow-derived M ϕ s were stained and analyzed for detection of total ROS using a Cellular ROS detection assay kit (Abcam, ab113851) following the manufacturer's instructions. Fluorescence intensity was detected using an FACSCanto II flow cytometer (BD Biosciences).

ELISA

Treated mouse bone marrow-derived M ϕ s and human PBMC-derived M ϕ s were subjected to ELISA. Total protein concentration was measured using a protein assay kit (Bio-Rad, 5000006). For detecting mouse and human PPAR γ activity, cells were analyzed using PPAR γ ELISA kits following the manufacturer's instructions (Biorbyt, orb775497 and Elabscience, E-EL-H1361, respectively). For measuring oxLDL level, cells were analyzed using a mouse oxLDL kit (Biorbyt, orb782036) and human oxLDL kit (Elabscience, E-EL-H6021). For analysis of mouse TNF and IL-1 β , cell lysis was analyzed using a mouse TNF ELISA Kit (Proteintech, KE10002) and a mouse IL-1 β ELISA kit (Proteintech, KE10003). Absorbance at 450 nm was detected with a Synergy H4 Hybrid microplate reader (BioTek). The PPAR γ activity and oxLDL level were calculated depending on the standard curve and normalized with total protein concentration.

TF activation array

Mouse BM-derived M ϕ s were irradiated at 5 Gy, followed by TF activity analysis with a TF activation profiling array (Signosis, FA-1102). In brief, cells were washed and lysed. The lysates were incubated with TF probe mix in TF-binding buffer. The assembly of TF-DNA complexes was isolated using an isolation column, followed by probe hybridization in a 96-well plate coated with probe sequences at 42 °C overnight. The signal of streptavidin-HRP (horseradish peroxidase) conjugates were measured using a Synergy H4 Hybrid microplate reader (BioTek).

Immunoblot

For cell sample preparation, cells were lysed with an NP-40 lysis buffer with a protease inhibitor cocktail (Roche, 11697498001). Total protein (20 μ g) was resolved by 4–

20% precast SDS–PAGE (Bio-Rad, 456-1094), followed by transfer. PVDF membranes were blotted with anti-arginase 1 antibody (1:500; Santa Cruz, sc-20150) and anti-glyceraldehyde-3-phosphate dehydrogenase antibody (1:3,000; Cell Signaling, 5174) at 4 °C overnight. Proteins were detected with HRP-conjugated secondary antibodies (Bio-Rad) and the bands were imaged by enhanced chemiluminescence development (GE Healthcare, RPN2232) using a Chemidoc Imager (Bio-Rad).

Immunofluorescence

Mouse tumor sections were subject to deparaffinization and rehydration and incubated with antigen retrieval solution (DAKO, S1699) for 20 min at 95 °C. Tissue sections were blocked with PBS containing 5% horse serum for 1 h and incubated with anti-GD2 antibody (1:200, BioLegend, 357302) at 4 °C overnight. After washing with PBS, sections were stained with Alexa Fluor 488-conjugated IgGs (1:500, Life Technologies) for 1 h at room temperature. Images were acquired using an Axio Imager microscope (Zeiss) equipped with an AxioCam 506 monochrome charge-coupled device camera (Zeiss).

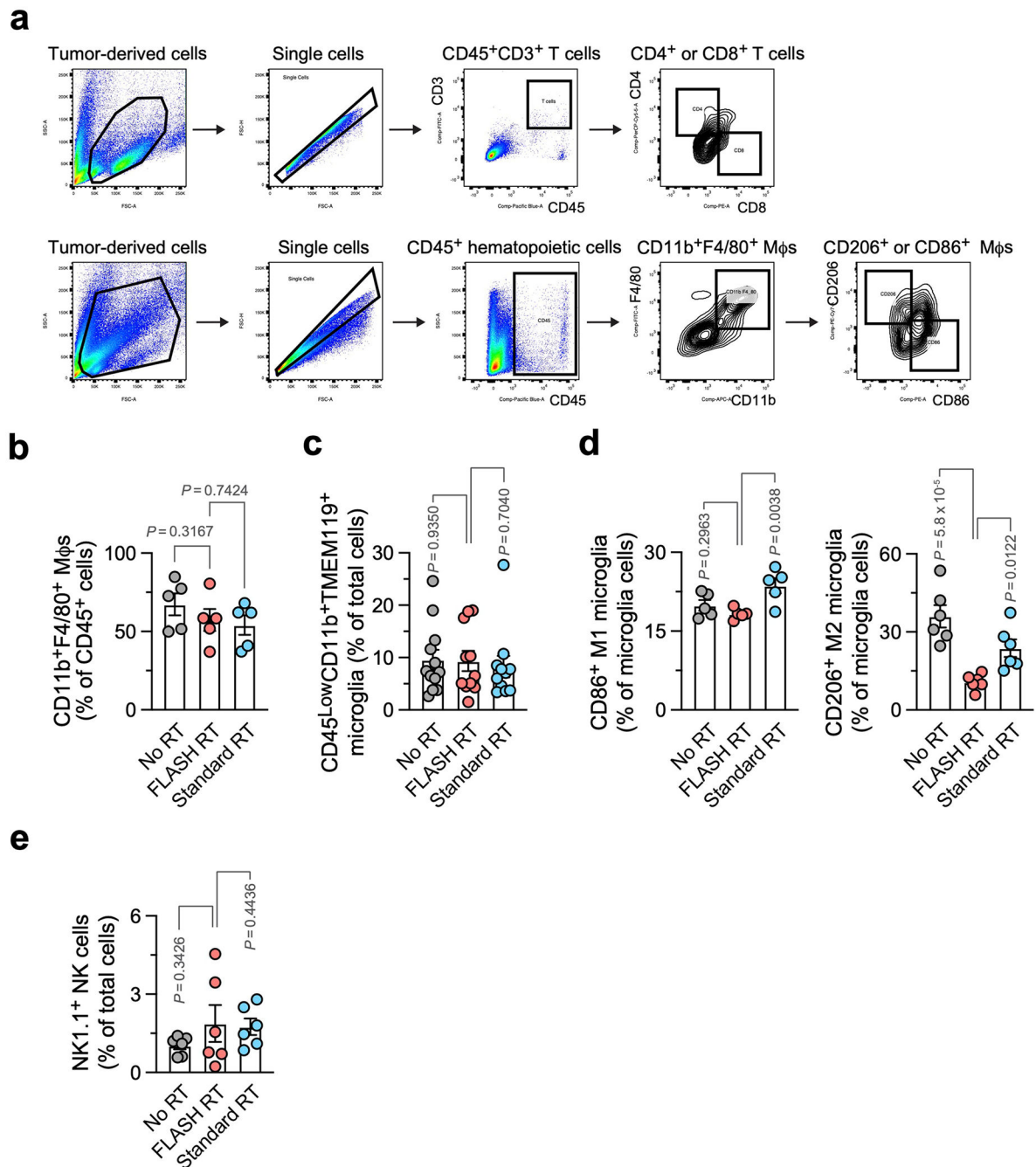
Statistics and reproducibility

All statistical tests were performed using Prism software (GraphPad, version 10.0). All statistical tests were two-sided. An unpaired Student's *t*-test was used to measure differences between the two groups. For multiple-group comparisons, one-way or two-way analysis of variance (ANOVA) was used to determine statistically significant differences between groups. Kaplan–Meier analysis with a log-rank test was performed for survival analysis. A *P* value lower than 0.05 was considered significant. The exact *P* value is shown in each figure. No statistical methods were used to predetermine sample sizes but we used adequate numbers of samples that would provide statistically significant results on the basis of our previous experience. The exact sample sizes are indicated in the figures or figure legends. To ensure the reproducibility of our results, all experiments were conducted with adequate replicates. All in vivo experiments were randomized to each experimental cohort. The investigators were not blinded to allocation during experiments or outcome assessments. Data distribution was assumed to be normal but this was not formally tested. No data were excluded from the analyses.

Reporting summary

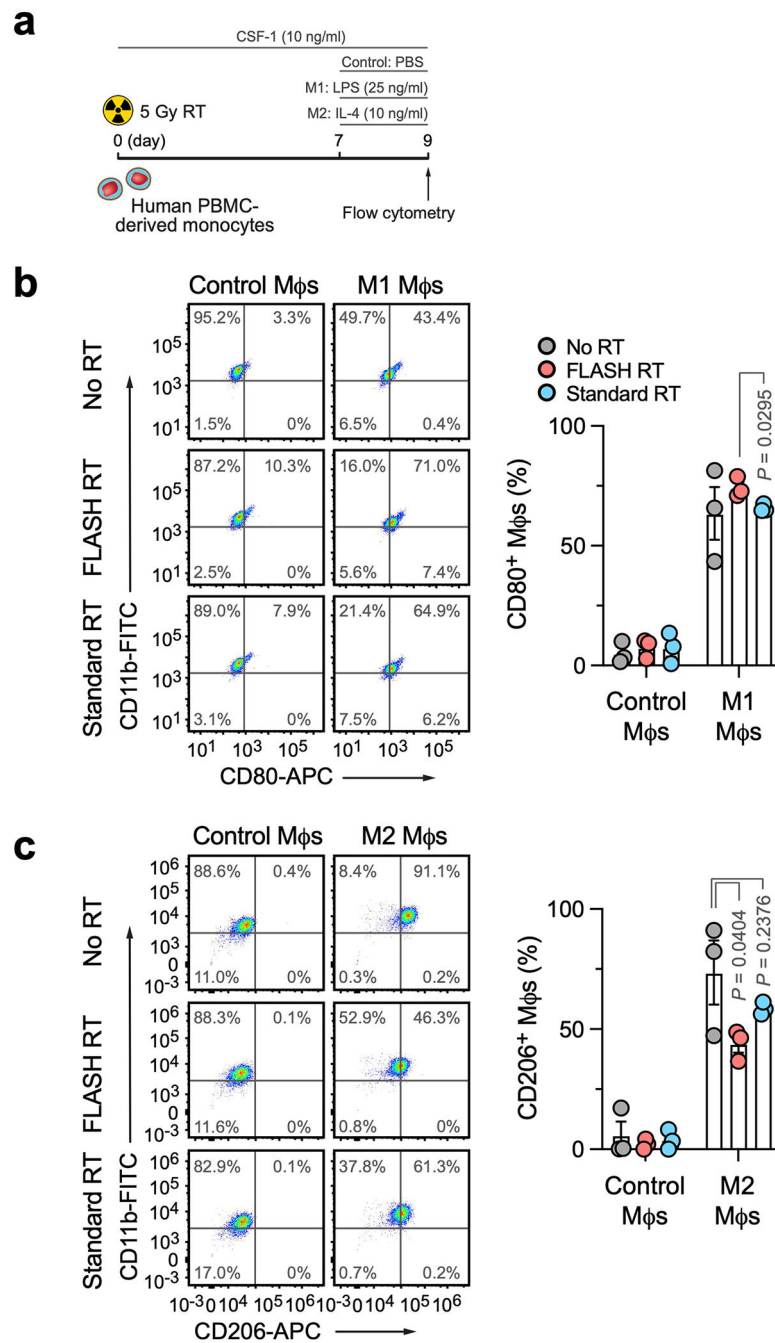
Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Extended Data

**Extended Data Fig. 1 | Effects of RT on tumor-associated Mfs, microglia, and NK cells.**

Medulloblastoma was genetically engineered in SmoM2 mice, followed by irradiation with FLASH or standard proton beam. Tumors were excised and subjected to flow cytometry analysis. **a**, Gating strategies for analysis of T cells and Mfs, corresponding to Fig. 1i–o. **b–e**, Analysis for **b**, CD11b⁺F4/80⁺ total Mfs (n = 5 mice), **c**, CD45^{Low}CD11b⁺TMEM119⁺ total microglia (n = 12 mice for no RT group, and n = 11 mice for FLASH and standard RT

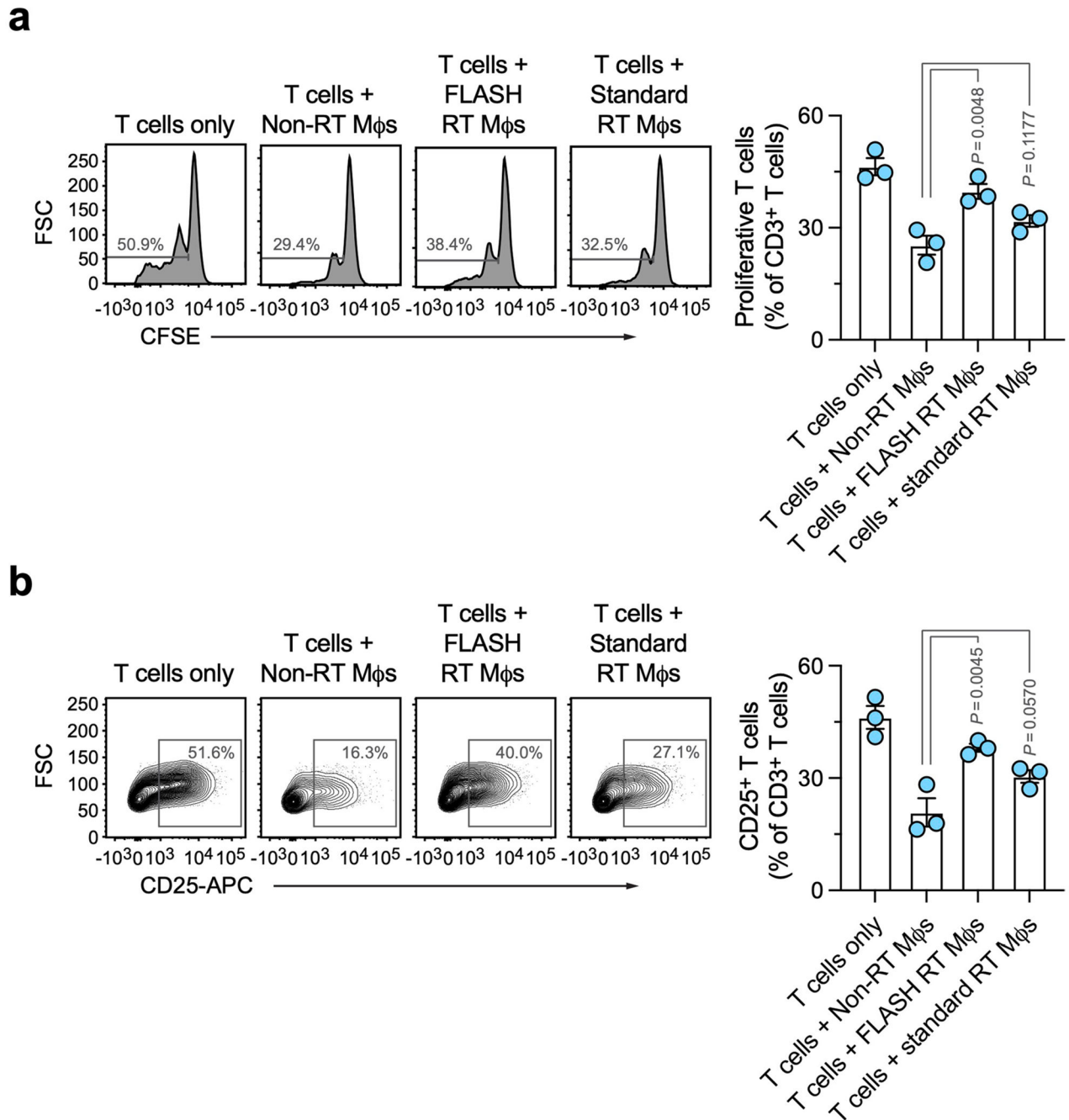
groups), **d**, CD86⁺ M1-like (n = 5 mice) and CD206⁺ M2-like microglia (n = 6 mice), and **e**, NK1.1⁺ NK cells (n = 6 mice). Statistical analysis by one-way ANOVA (mean ± SEM).



Extended Data Fig. 2 | Effects of RT on human Mφ polarization in vitro.

Human PBMC-derived Mφs were irradiated with FLASH or standard proton beam, followed by treatment with LPS or IL-4. **a**, Experimental procedure. **b,c**, After treatment with **b**, LPS or **c**, IL-4, cells were analyzed by flow cytometry. Left, representative cell sortings.

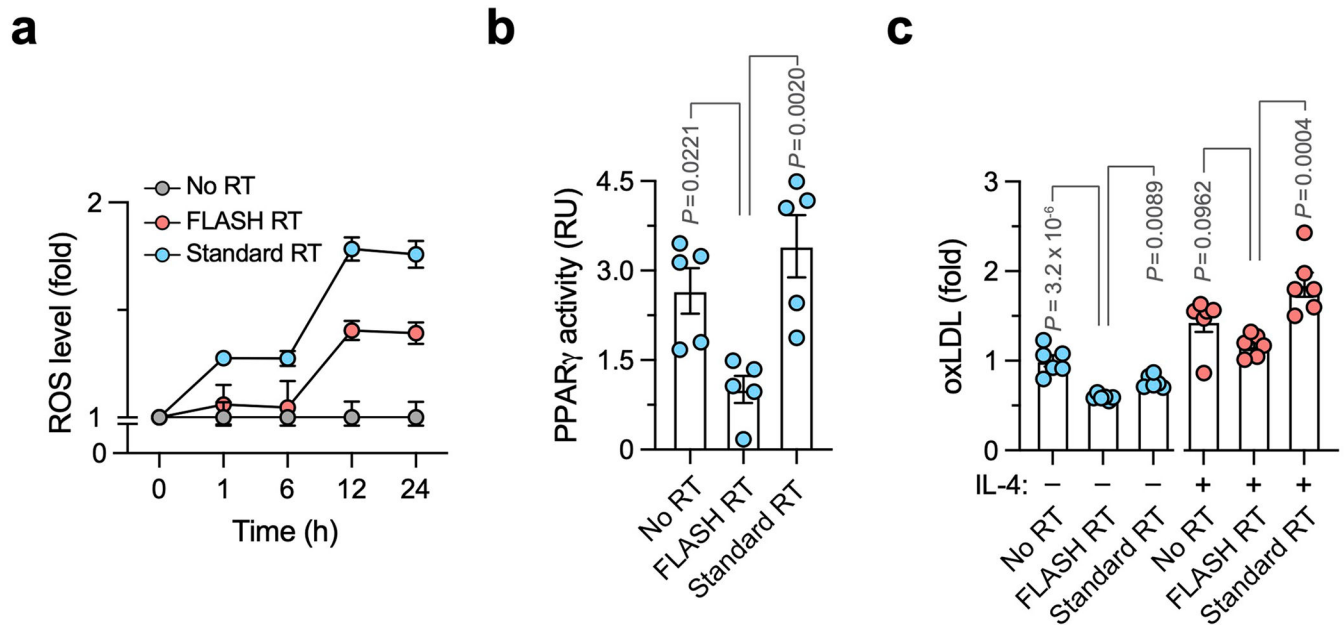
Right, quantified results ($n = 3$ human participants, mean $\pm$ SEM). **b**, Statistical analysis by two-tailed Student's *t* test. **c**, Statistical analysis by one-way ANOVA.



Extended Data Fig. 3 |. Effects of irradiated Mφs on T cell functions in vitro.

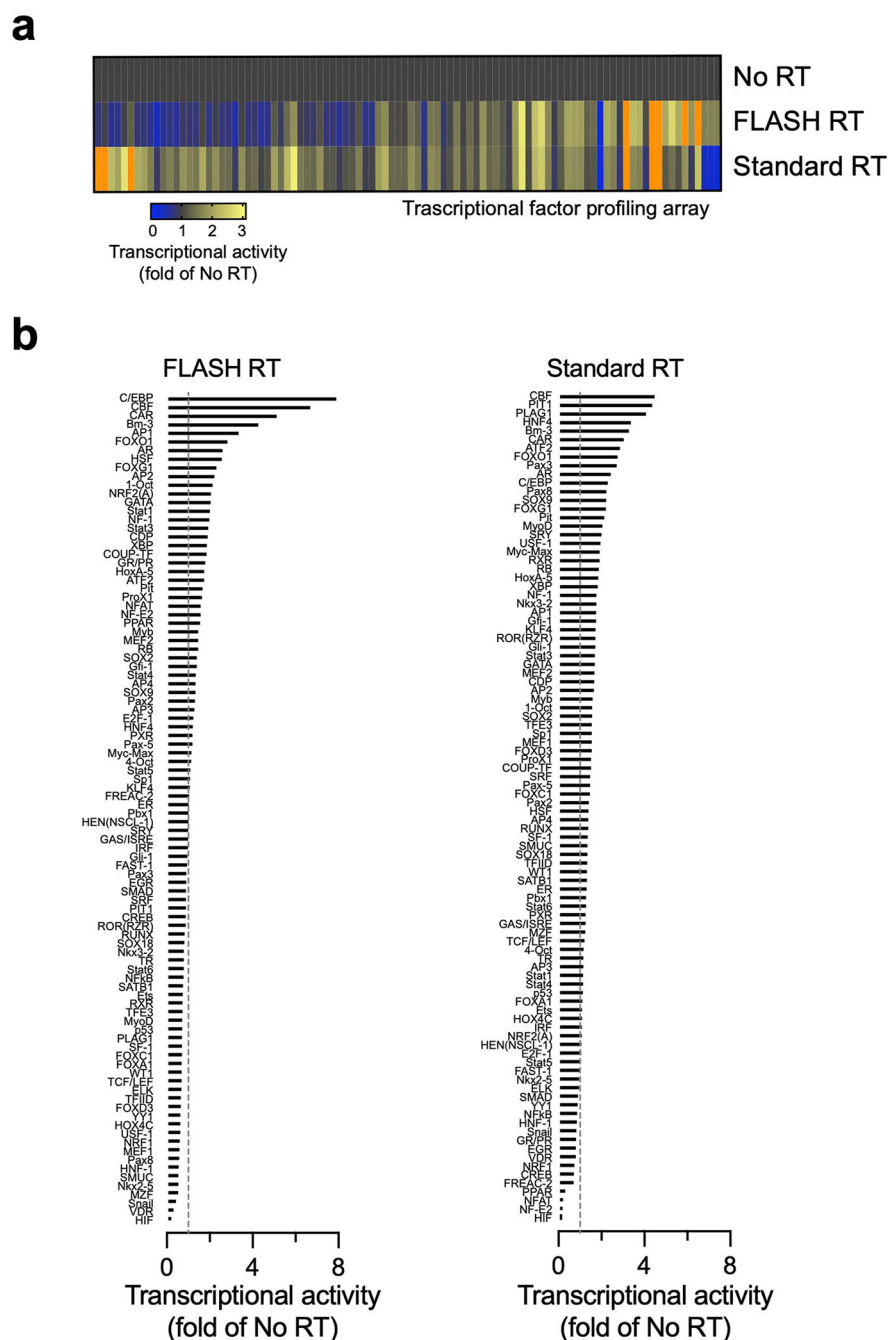
Human PBMC-derived Mφs cells were irradiated by FLASH or standard proton beam and treated with IL-4 for 2 days. Human PBMC-derived CD3⁺ T cells were stimulated with CD3/CD28 beads for 3 days, and loaded with CFSE. Treated Mφs and T cells were incubated for 2 days, followed by flow cytometry analysis. **a**, CFSE was analyzed in CD3⁺

T cells. Left, representative cell sortings. Right, quantified results (n = 3 human participants, mean $\pm$ SEM). Statistical analysis by one-way ANOVA. **b**, CD25 expression was analyzed in CD3⁺ T cells. Left, representative cell sortings. Right, quantified results (n = 3 human participants, mean $\pm$ SEM). Statistical analysis by one-way ANOVA.



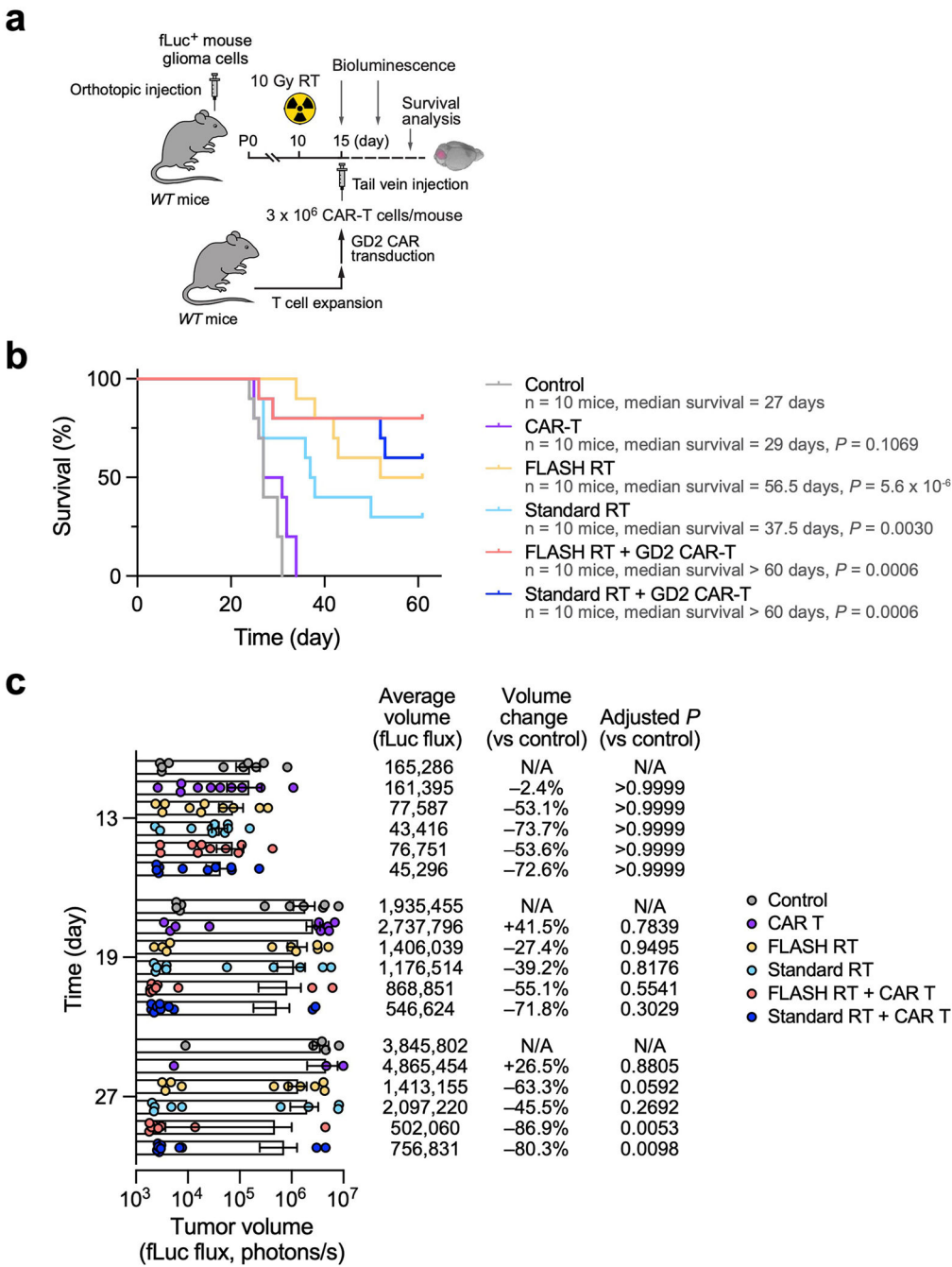
Extended Data Fig. 4 |. Effects of RT on ROS generation, PPAR γ activity and oxLDL production in human M ϕ s.

Human PBMC-derived M ϕ s were irradiated with FLASH or standard proton beam. **a**, Total ROS were analyzed at different time post-irradiation (mean $\pm$ SEM, n = 3 human participants). **b**, PPAR γ activity was measured 24 h after irradiation (mean $\pm$ SEM, n = 5 human participants). Statistical analysis by one-way ANOVA. **c**, Human PBMC-derived M ϕ s were irradiated with FLASH or standard proton beam, followed by treatment with or without IL-4. Cell lysates were subjected to oxLDL analysis (mean $\pm$ SEM, n = 6 human participants). Statistical analysis by one-way ANOVA.



Extended Data Fig. 5 | Effects of RT on transcriptional factor activity in vitro.

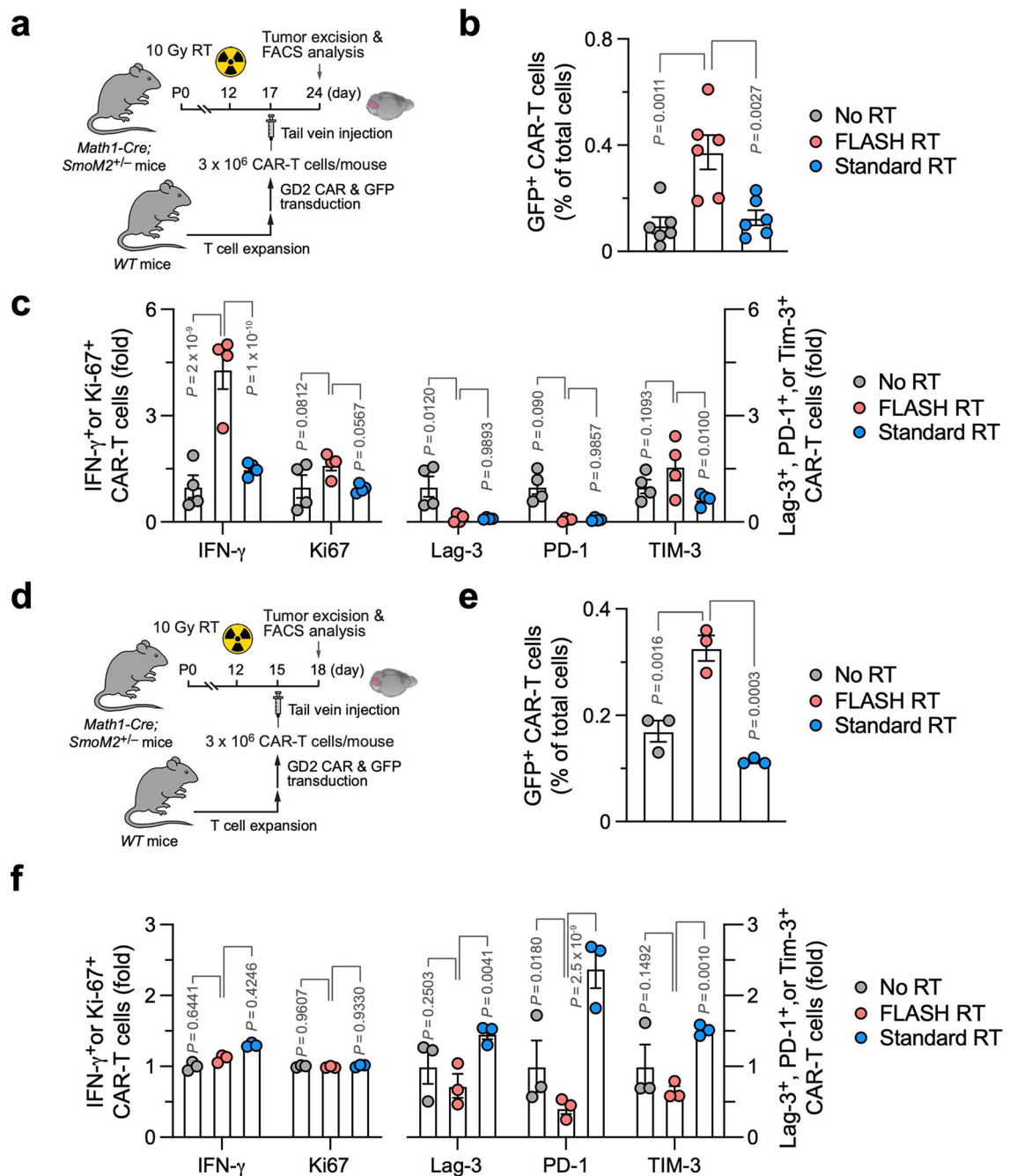
Mouse BM-derived Mφs (pooled from 3 mouse samples for each group) were irradiated with FLASH or standard proton beam, and subjected to analysis with a transcriptional factor profiling assay. The activity of 96 transcriptional factors was expressed as the fold of no RT group. **a**, Heatmap. **b**, Ranked activity.



Extended Data Fig. 6 |. Combination of RT with CAR T cell therapy in a syngeneic mouse glioma model.

Glioma was induced in mice by orthotopic transplantation with GL261 mouse glioma cells, followed by FLASH or standard RT and GD2 CAR-T cell therapy. **a**, Experimental procedures. **b**, Animal survival was monitored for 60 days (n = 10 mice). Statistical analysis by a two-tailed Log-rank Mantel-Cox test. **c**, Tumor volume was measured by bioluminescence imaging (n = 10 mice, mean ± SEM). Note: after standard RT plus CAR T

cell treatment, one mouse developed neurological symptoms at late day 26 and was imaged at day 27. Statistical analysis by two-way ANOVA.



Extended Data Fig. 7 | Effects of RT on CAR T infiltration and activity in vivo.

a-c, 5 days after irradiation, SmoM2 mice were treated with GD2 CAR-T cells. **a**, Experimental procedures. **b,c**, Tumors were excised 7 days after CAR-T cell therapy, followed by flow cytometry analysis. (**b**, Analysis of GFP⁺ CAR-T cells ($n = 6$ mice, mean $\pm$ SEM). Statistical analysis by one-way ANOVA. **c**, Analysis of IFN- γ ⁺, Ki-67⁺, Lag-3⁺,

PD-1⁺ and Tim-3⁺ GFP⁺ CAR T cells (n = 4 mice, mean ± SEM). Statistical analysis by two-way ANOVA. **d-e**, 3 days after irradiation, SmoM2 mice were treated with GD2 CAR-T cells. **d**, Experimental procedures. **e,f**, Tumors were excised 3 days after CAR-T cell therapy, followed by flow cytometry analysis. **e**, Analysis of GFP⁺ CAR T cells (n = 3 mice, mean ± SEM). Statistical analysis by one-way ANOVA. **f**, Analysis of IFN-γ⁺, Ki-67⁺, Lag-3⁺, PD-1⁺ and Tim-3⁺ GFP⁺ CAR T cells (n = 3 mice, mean ± SEM). Statistical analysis by two-way ANOVA.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data availability

Single-cell and bulk RNAseq data were deposited to the National Center for Biotechnology Information's Gene Expression Omnibus under accession numbers [GSE246970](#) and [GSE246969](#), respectively. All remaining data are available within the article and the Supplementary Information or available from the authors upon request. Source data are provided with this paper.

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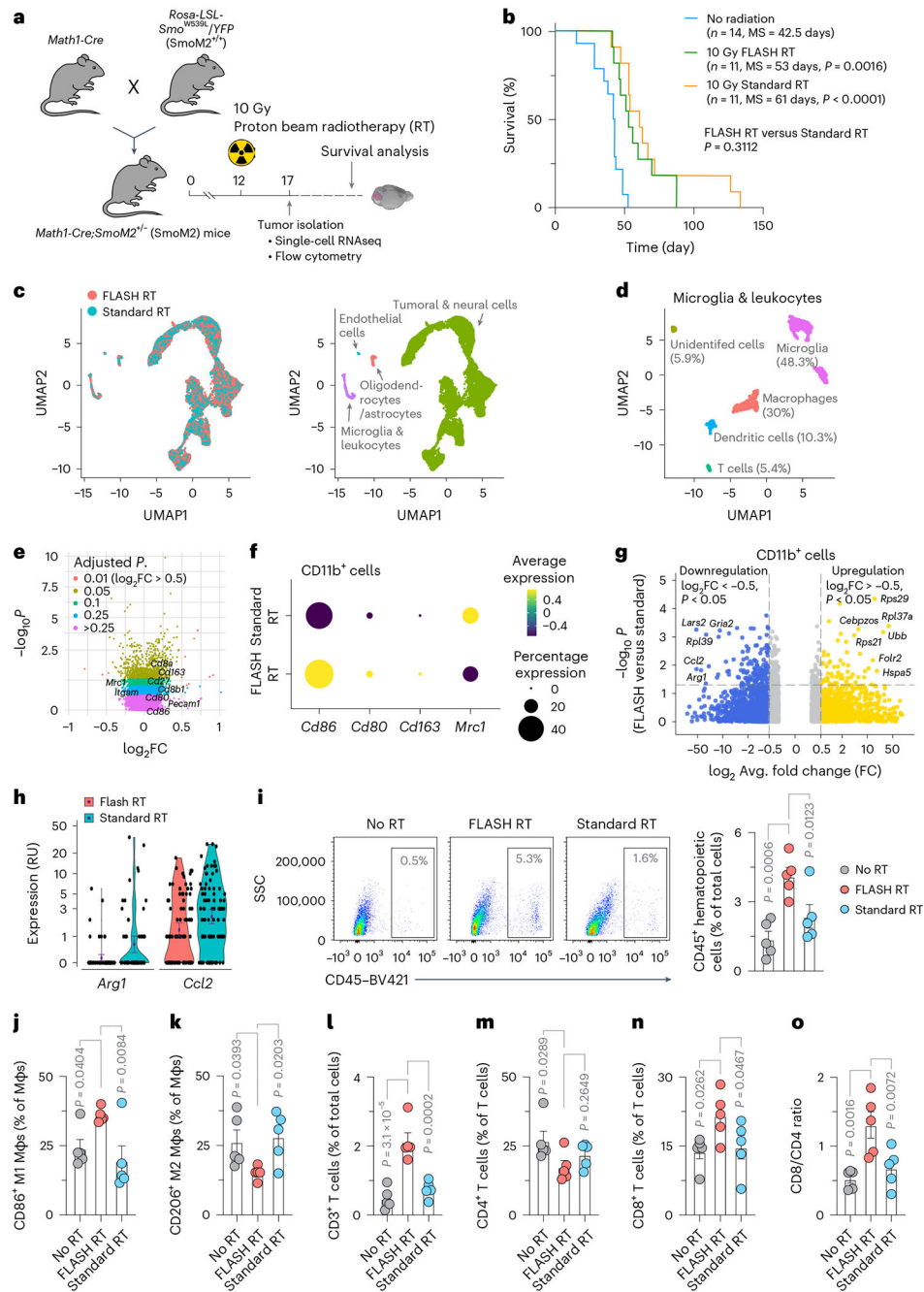


Fig. 1 | FLASH RT shows antitumor effects and skews Mφs toward a proinflammatory phenotypes in mouse MB.

MB was genetically engineered in SmoM2 mice, followed by irradiation with FLASH or standard proton beam. **a**, Experimental procedure. **b**, After irradiation with a 10-Gy proton beam, animal survival was monitored ($n = 11-14$ mice; specific n value of each group listed in the figure). Statistical analysis by log-rank test. **c-h**, Single-cell RNAseq analysis (pooled from $n = 3$ mice for each group). **c**, UMAP analysis of transcriptome gene signature in all tumor-derived cells. Left, integration of RNAseq data from samples irradiated with

FLASH or standard proton beam. Right, distribution of cell clusters. **d**, UMAP analysis of transcriptome gene signature in microglia and leukocytes. **e**, Top regulated genes in microglia and leukocytes. FC, fold change. **f**, CD86, CD80, CD163 and CD206 (*Mrc1*) expression in CD11b⁺ myeloid cells. **g**, Top altered genes in CD11b⁺ myeloid cells. **h**, Arginase 1 (*Arg1*) and MCP1 (*Ccl2*) expression in CD11b⁺ myeloid cells. **i–o**, Flow cytometry analysis for CD45⁺ hematopoietic cells (**i**), CD86⁺ M1-like Mφs (**j**) and CD206⁺ M2-like Mφs (**k**) in CD11b⁺F4/80⁺ Mφs, CD45⁺CD3⁺ T cells in total cells (**l**) and CD4⁺ (**m**) and CD8⁺ (**n**) T cells in CD45⁺CD3⁺ T cells and ratio of CD8⁺/CD4⁺ T cells (**o**) ($n = 5$ mice, mean $\pm$ s.e.m.). Statistical analysis by one-way ANOVA. **i**, Left, representative cell sortings. Right, quantified results.

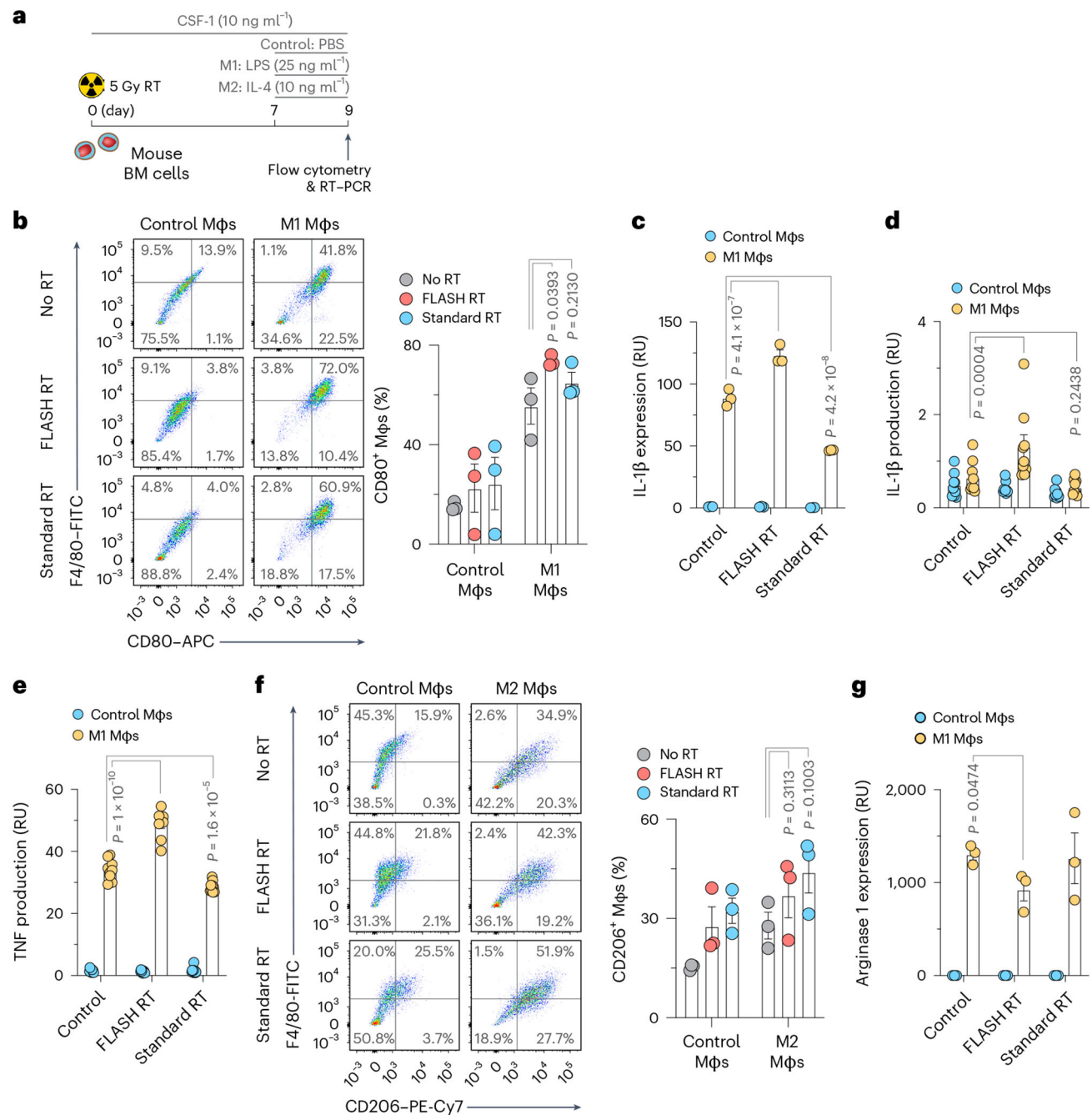


Fig. 2 | FLASH radiation stimulates the capacity for stimulus-dependent proinflammatory polarization in Mφs.

Mouse BM-derived Mφs were irradiated with FLASH or standard proton beam, followed by treatment with LPS or IL-4. **a**, Experimental procedure. **b–e**, After treatment with LPS, cells were analyzed by flow cytometry (**b**), RT-PCR (**c**) or ELISA (**d,e**). **b**, Left, representative cell sortings. Right, quantified results ($n = 3$ mice, mean $\pm$ s.e.m.). **c**, Quantified results of RT-PCR ($n = 3$ mice, mean $\pm$ s.e.m.). **d**, ELISA results for IL-1β expression ($n = 9$ mice, mean $\pm$ s.e.m.). **e**, ELISA results for TNF expression ($n = 9$ mice, mean $\pm$ s.e.m.). **b–e**,

Statistical analysis by one-way ANOVA. **f,g**, After treatment with IL-4, cells were analyzed by flow cytometry (**f**) and RT-PCR (**g**). **f**, Left, representative cell sortings. Right, quantified results ($n = 3$ mice, mean $\pm$ s.e.m.). Statistical analysis by one-way ANOVA. **g**, Quantified results of RT-PCR ($n = 3$ mice, mean $\pm$ s.e.m.). Statistical analysis by two-tailed Student's t -test.

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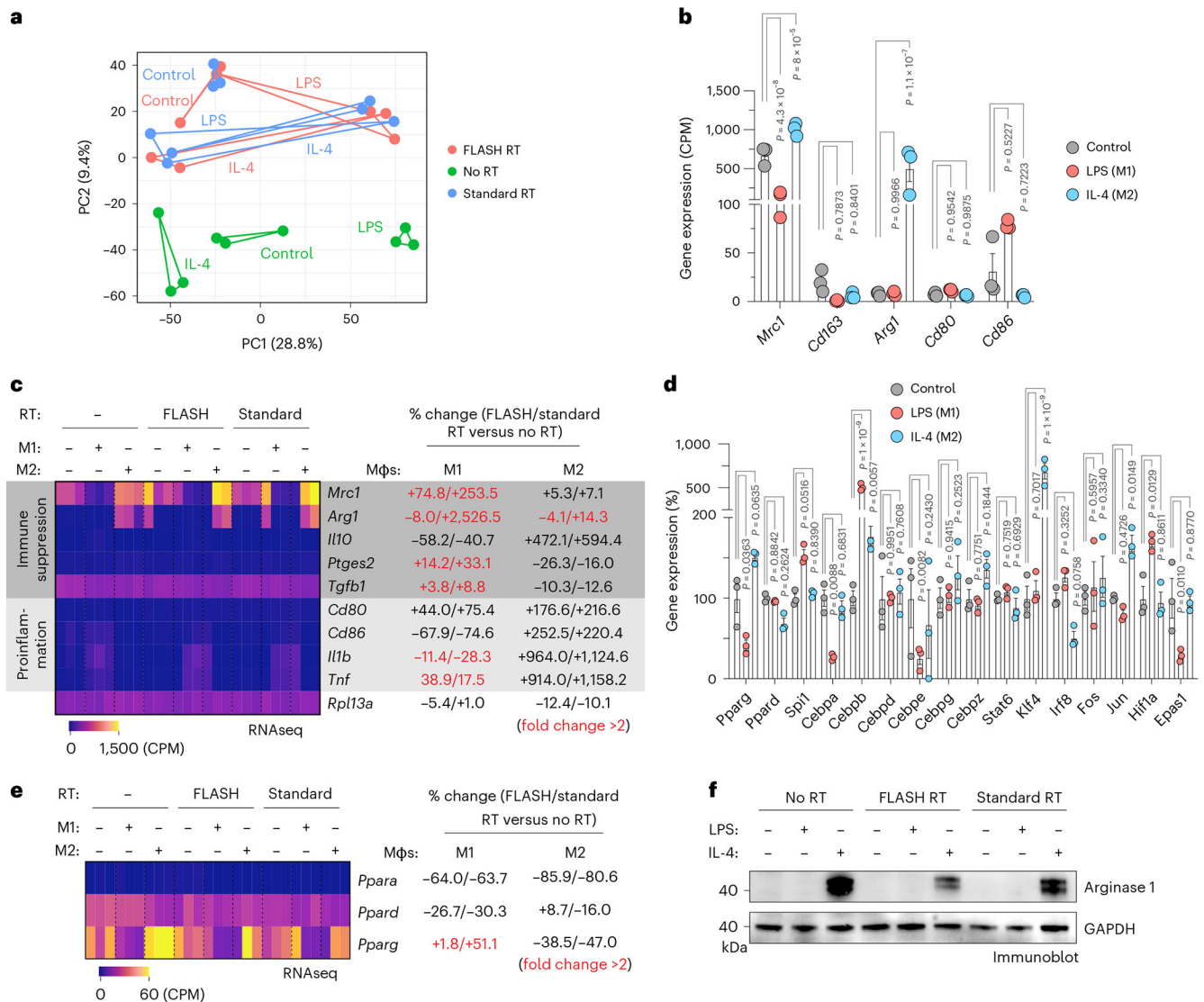


Fig. 3 | FLASH radiation induces less PPAR γ expression and immunosuppressive phenotypes in M ϕ s.

Mouse BM-derived M ϕ s were irradiated with 5-Gy FLASH or standard proton beam, followed by treated with LPS (M1) or IL-4 (M2). **a–e**, RNA was extracted and analyzed by RNAseq ($n = 3$ samples per group, pooled from three mice). **a**, Principal component analysis of all of the mapped genes. **b**, Expression of M1 and M2 marker genes in unirradiated M ϕ s treated with or without LPS or IL-4 (mean $\pm$ s.e.m.). Statistical analysis by two-way ANOVA. **c**, Expression of immunosuppression-associated and proinflammation-associated genes in irradiated M1 or M2 M ϕ s. Left, heat map of gene expression. Right, quantified changes over unirradiated cells (average percentage change). **d**, Expression of M2 polarization-associated TFs in unirradiated M ϕ s treated with or without LPS or IL-4 (mean $\pm$ s.e.m.). Statistical analysis by one-way ANOVA. **e**, Expression of PPAR TFs in irradiated control, M1 M ϕ s or M2 M ϕ s. Left, heat map of gene expression. Right, quantified

changes over unirradiated cells (average percentage change). **f**, Treated cells were analyzed by immunoblot. This experiment was repeated independently twice with similar results.

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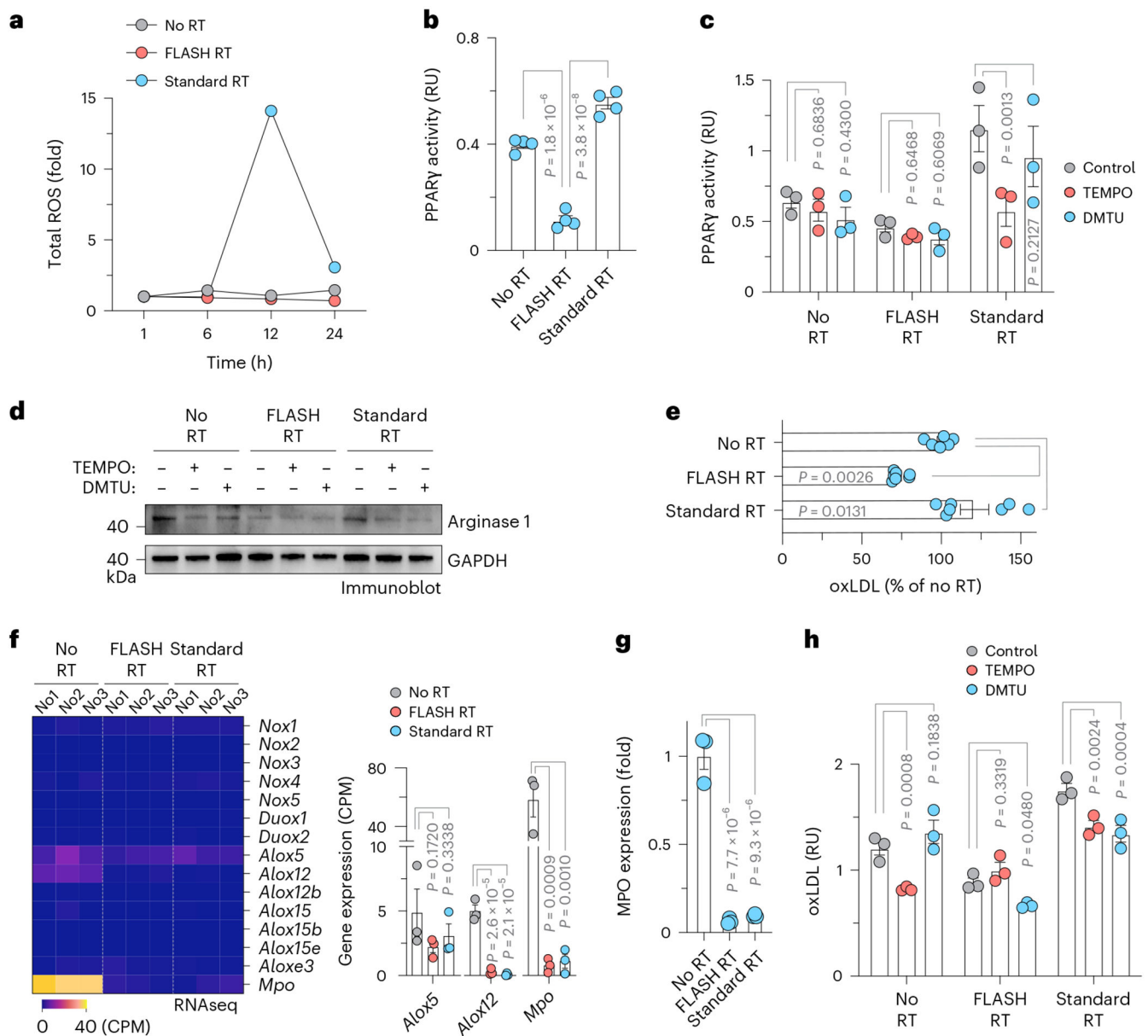


Fig. 4 | FLASH radiation inhibits oxidase expression and oxLDL generation to reduce PPAR γ activity and arginase 1 expression, while standard radiation induces redox-dependent PPAR γ activation and arginase 1 expression in M ϕ s.

a,b, Mouse BM-derived M ϕ s were irradiated with 5-Gy FLASH or standard proton beam.

a, Total ROS were analyzed at different times after irradiation (mean $\pm$ s.e.m., $n = 4$ mice). **b**, PPAR γ activity was measured 24 h after radiation (mean $\pm$ s.e.m., $n = 4$ mice).

Statistical analysis by one-way ANOVA. **c,d**, Mouse M ϕ s were treated with TEMPO or DMTU, followed by FLASH or standard irradiation. **c**, PPAR γ activity was measured 24 h after radiation (mean $\pm$ s.e.m., $n = 4$ mice). Statistical analysis by two-way ANOVA.

d, Cell lysate was immunoblotted. This experiment was repeated independently twice with similar results.

e-g, Mouse M ϕ s were irradiated with 5-Gy FLASH or standard radiation. **e**, Cell lysates were subject to oxLDL analysis (mean $\pm$ s.e.m., $n = 7$ mice, pooled from two

experiments). Statistical analysis by one-way ANOVA. **f**, RNA was extracted and analyzed by RNAseq ($n = 3$ samples, pooled from three mice; total of 27 mice). Left, heat map of oxidase gene expression. Right, quantified results. Statistical analysis by two-way ANOVA. **g**, RNA was extracted and analyzed by RT-PCR (mean $\pm$ s.e.m., $n = 3$ mice). Statistical analysis by one-way ANOVA. **h**, Mouse BM-derived M ϕ s were treated with TEMPO or DMTU, followed by FLASH or standard radiation. Cell lysates were subjected to oxLDL analysis (mean $\pm$ s.e.m., $n = 3$ mice). Statistical analysis by one-way ANOVA.

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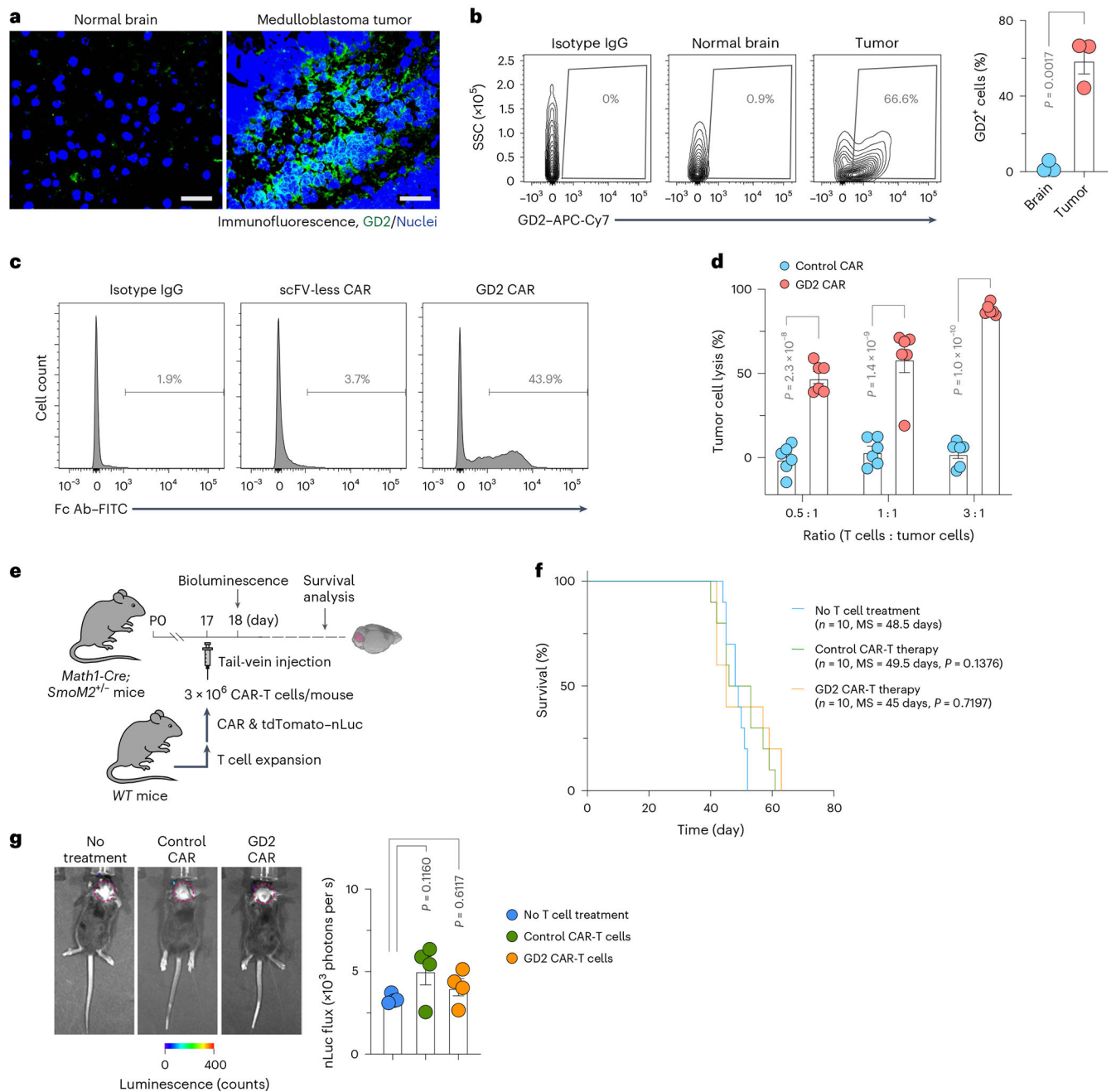


Fig. 5 | GD2 CAR-T cells show robust activity in vitro but minimal therapeutic efficacy in vivo. **a,b**, Normal brain and tumor tissues were excised from WT and SmoM2 mice. **a**, Tissue sections were immunostained with anti-GD2 antibody ($n = 4$ mice). Representative images are shown. Scale bars, 50 μ m. **b**, Tissue-derived single-cell suspensions were immunostained and analyzed by flow cytometry. Left, representative sortings. Right, quantified results ($n = 3$ mice, mean $\pm$ s.e.m.). Statistical analysis by two-tailed Student's *t*-test. **c–g**, GD2 CAR-T therapy in mice. **c**, Preparation of CAR-T cells. Mouse T cells were retrovirally transduced to express control or GD2 CAR. Representative sortings are shown, **d**, Mouse tumor cells were isolated from MB tumors and incubated with control or GD2 mouse CAR-T cells,

followed by cell lysis assay ($n = 6$ assays, mean $\pm$ s.e.m.). Statistical analysis by two-way ANOVA. **e–g**, SmoM2 mice were treated with or without control or GD2 CAR-T cells. **e**, Experimental procedures. **f**, Animal survival was monitored. Statistical analysis by log-rank test. **g**, Mice were imaged by bioluminescence. Left, representative images. Dashed circles indicate the brain area. Right, quantified luminescence signals in the brain area ($n = 4$ mice, mean $\pm$ s.e.m.). Statistical analysis by one-way ANOVA.

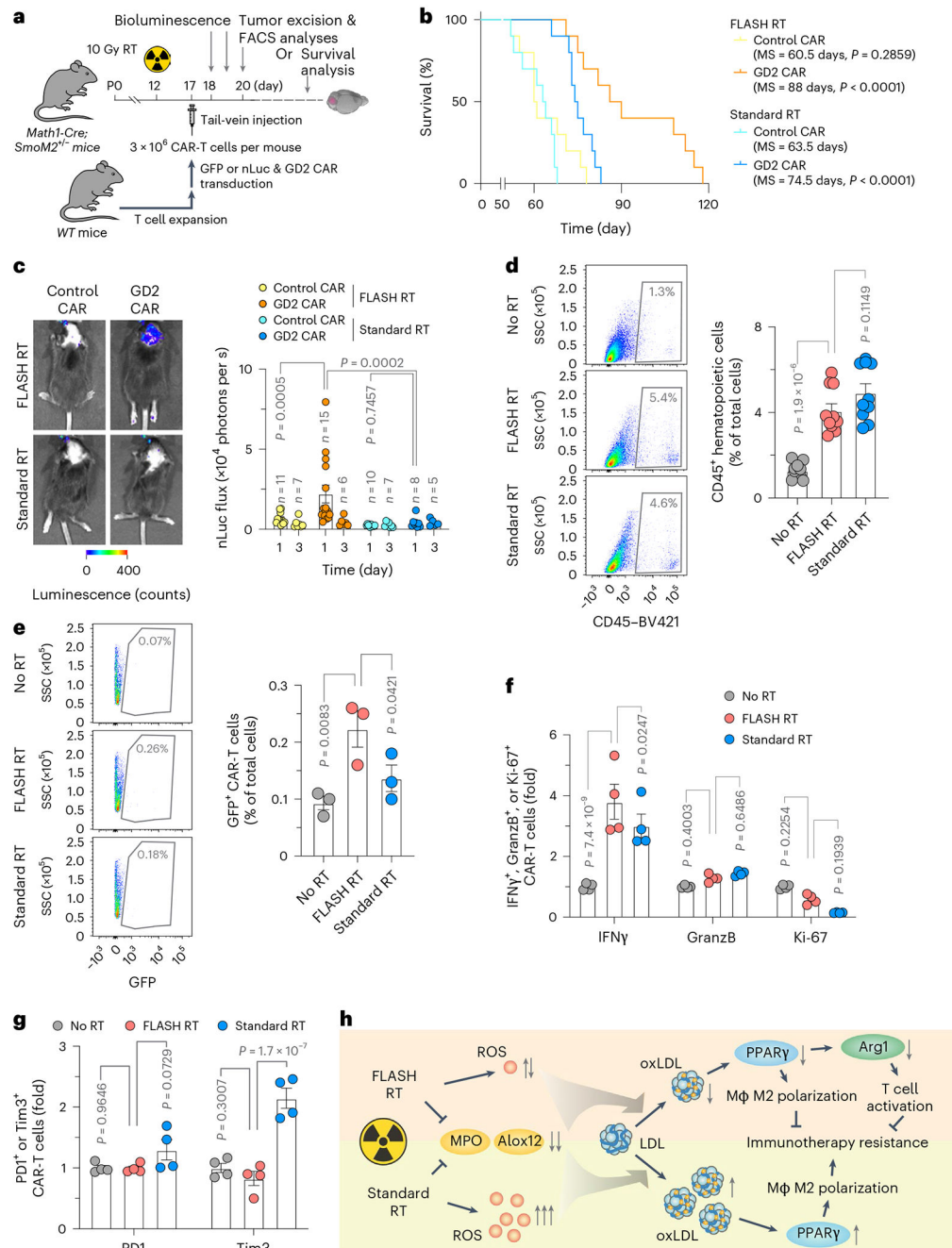


Fig. 6 | FLASH RT overcomes MB resistance to GD2 CAR-T cell immunotherapy.

SmoM2 mice were irradiated and treated with control or GD2 CAR-T cells. **a**, Experimental procedures. **b**, Animal survival was monitored ($n = 10$ mice). Statistical analysis by log-rank test. **c**, CAR-T cell infiltration was imaged by bioluminescence. Left, representative images. Right, quantified results on days 1 and 3 after CAR-T cell injection ($n = 5$ – 15 mice; specific n value of each group listed in the figure; mean $\pm$ s.e.m.). Statistical analysis by two-way ANOVA. **d–g**, Tumors were excised 3 days after CAR-T cell therapy, followed by flow cytometry analysis. **d**, CD45⁺ hematopoietic cells. Left, representative cell sortings. Right,

quantified results ($n = 10$ mice, mean $\pm$ s.e.m.). Statistical analysis by one-way ANOVA. **e**, GFP⁺ CAR-T cells. Left, representative cell sortings. Right, quantified results ($n = 3$ mice, mean $\pm$ s.e.m.). Statistical analysis by one-way ANOVA. **f**, IFN γ ⁺, GranzB⁺ and Ki67⁺ GFP⁺ CAR-T cells. Quantified results ($n = 4$ mice, mean $\pm$ s.e.m.). Statistical analysis by two-way ANOVA. **g**, PD1⁺ and Tim3⁺ GFP⁺ CAR-T cells. Quantified results ($n = 4$ mice, mean $\pm$ s.e.m.). Statistical analysis by two-way ANOVA. **h**, A schematic model. FLASH radiation abrogates expression of oxidases including Mpo and Alox12 and slightly stimulates ROS generation in M ϕ s, culminating in decreases in oxLDL production and PPAR γ activation, which in turn increases T cell activity through reduced arginase 1 (*Arg1*) expression and enhances M ϕ M1-like proinflammatory polarization, eventually overcoming tumor resistance to T cell-based cancer immunotherapy. Standard RT also abolishes oxidase expression but robustly enhances ROS production in M ϕ s, resulting in increases in oxLDL production and PPAR γ activation, which drives M ϕ M2-like anti-inflammatory polarization and induces tumor resistance to immunotherapy.